

Remote Work, Wages, and Competition: Disentangling Demand and Supply Influences*

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Abstract

This paper examines the impact of remote work on wages by modeling the labor market as an oligopsonistic network where firms strategically set wages and modalities. Using 2 million job postings and a BERT classification pipeline, we distinguish between fully remote and hybrid roles. To solve the reflection problem in our competition model, we use the behavior of competitors of competitors as instruments, leveraging the bipartite application network. We find a 24% wage premium for fully remote work, suggesting a loss of local monopsony power, while hybrid roles carry a 3.6% penalty. Remote work expands the reach of the market while inducing high-quality applicant sorting.

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1 Introduction

The global shift to remote work is an important structural transformation of the labor market in recent history. Although a growing body of research has documented the rapid adoption of work-from-home (WFH) and hybrid (HYB) arrangements following the COVID-19 pandemic, the consequences for wage determination and market structure are not fully understood. Standard economic forces imply ambiguous predictions about the impact of remote work on compensation. On the supply side, a compensating differentials framework suggests that workers should accept lower wages in exchange for the amenity value of flexibility, reduced commute costs, and better work-life balance. On the demand side, firms might pay higher wages for remote-capable roles to attract top talent from a broader geographic pool, capitalize on potential productivity gains, or offset the loss of local monopsony power. Empirical evidence has yet to definitively disentangle these competing influences.

Our analysis provides a resolution to this question by demonstrating that fully remote (WFH) and hybrid (HYB) work are not merely different degrees of the same phenomenon but distinct strategic choices with distinct impacts on the labor market. Using a comprehensive data set of more than two million job postings from the Chilean job board trabajando.com, we find that fully remote work is associated with a substantial wage premium of approximately 24% in our preferred instrumental variable specifications. This finding suggests that as firms move to fully remote models, they lose their local monopsony power and must compete in a larger, geographically neutral market. In contrast, hybrid arrangements carry a relatively small wage penalty—around 3.6%—reflecting the fact that these markets remain geographically constrained, allowing the non-pecuniary value of the amenity to predominate in the wage-setting process. Furthermore, we show that remote modalities significantly expand the labor market’s reach, increasing the area of a catchment, the coverage of the population, and the number of municipalities from which firms draw talent. However, we also document a “portfolio effect,” a theoretically expected diversification of the inherent risk of applications when job seekers can apply for multiple ads simultaneously ([Galenianos and Kircher, 2009](#)). In a nutshell, an exogenous wage increase for focal ad lead to more intense competition for that position and higher-quality applicant sorting. In response to this change, other competitor ads strategically differentiate by offering less attractive wages, and thus less intense competition to attract additional applications.

These empirical findings are made possible by two primary contributions. Methodologically, we develop a multi-stage classification pipeline using a fine-tuned Spanish-language BERT model (BETO) to identify WFH and HYB job ads from unstructured text. Using unique features of the trabajando.com dataset, we are able to observe wage posted by firms

even when they are hidden from applicants. This allows us to circumvent the sample selection biases inherent in most online job board data, where wage information is often missing for a non-random subset of postings. (Batra et al., 2023). Furthermore, by using text analysis to identify full-time status, we ensure that our comparisons are based on homogeneous work-hour arrangements, isolating the effect of remote work from the confounding factor of total hours worked.

Theoretically and empirically, we move beyond the assumption of a Walrasian labor market by modeling the environment as oligopsonistic. We posit that firms’ decisions regarding wages and work modalities are strategic reactions to the behavior of their competitors. To address the endogeneity of these decisions, we leverage the bipartite network structure formed by job applications. By linking advertisements through the common pool of workers who apply to them, we identify ad-specific local labor markets. To solve the “reflection problem” often found in social-effects models, we generate instruments based on the characteristics of the competitors of the firm’s own competitors—what we refer to as degree-2 network averages. This strategy allows us to isolate exogenously induced shifts in modalities and wages to estimate causal effects under sensible assumptions.

The paper is organized as follows. Following this introduction, the second section provides a review of related literature, situating our work within the broader context of research on remote work, productivity, and the use of text-as-data in labor economics. We discuss the evolving consensus on remote work productivity and the heterogeneous valuations workers place on flexibility. The third section describes in detail the data from *trabajando.com*. and discusses the unique value of having access to both firm-posted wages and applicant-expected wages, as well as the rich demographic information available for job seekers.

In the fourth section, we detail our core methodology for identifying WFH and work-hour arrangements. We describe the construction of our training dataset, the recruitment and training of human annotators, and the multi-round process used to define granular remote work categories. We demonstrate that the BETO model achieves high accuracy in classifying these modalities and that our results are robust to different data-curation choices. The fifth section introduces our econometric framework and the application of network-based social effect models to the labor market. We define the structural equations for wages and modalities and formalize our identification strategy using the application network.

The sixth section presents our main empirical results. We first examine wage behavior, contrasting the OLS and IV estimates to highlight the importance of accounting for endogeneity and strategic interactions. We then turn to modality offers, showing how the prevalence of WFH or hybrid work among competitors directly influences an employer’s own posting decisions. Finally, we explore the mechanisms of competition and the characteristics

of the applicants. We present our findings on how remote work expands the geographic size of the labor market and how it affects the sorting of applicants based on their wage expectations and educational attainment. We also provide robustness checks by re-estimating our models for the post-pandemic subsample to determine if a structural change occurred.

In summary, this paper provides a novel view of the labor market that is particularly relevant after the massive technological and social changes triggered by COVID-19. By combining advanced NLP techniques with a structural network approach, we demonstrate that the transition to remote work is not just a change in office location, but a fundamental redefinition of how firms and workers compete. Our findings on the WFH wage premium and the hybrid wage penalty suggest that the geography of a job continues to be a primary determinant of market power and compensation, even in an increasingly digital world.

2 Related Literature

Our paper is related to several strands of literature. As other contributors, we also report an abrupt surge in WFH using online job ads due to COVID-19 and a subsequent adoption of these work arrangements in the aftermath of the pandemic (Bick et al., 2023). The pandemic prompted a collective “learning experiment,” changing established managerial views on remote work (Barrero et al., 2021; Erdsiek and Rost, 2023). In this new stable equilibrium with a high prevalence of work from home (WFH), its effect on wages remains a relatively unexplored subject.

From the labor supply perspective, WFH is a valued, non-pecuniary job amenity for which a compensating differential exists. Hence, workers are willing to trade wages for flexibility. Foundational pre-pandemic work by Mas and Pallais (2017) estimated this trade-off, finding that workers, on average, were willing to forgo approximately 8% of their wages for a WFH option. More recent estimates of this willingness-to-pay range from 7.7% among German employees (Nagler et al., 2024) to 25% for U.S. tech workers (Cullen et al., 2025). This valuation is not homogeneous; women, for example, appear to have a stronger preference for flexibility (Banerjee et al., 2025). Intriguingly, while WFH reduces the gender gap in willingness-to-pay to avoid commuting, it does not close it (Nagler et al., 2024). On a macroeconomic scale, Barrero et al. (2022) show that this widespread amenity has moderated inflation through a potential 2.0 percentage point reduction in nominal wage growth pressures over two years.

However, the demand-side story is less clear. A firm’s wage offer should be linked to productivity, yet the evidence here remains notoriously ambiguous. The well-known RCT by Bloom et al. (2015) indicates a productivity gain of 13% for routine call-center tasks. In con-

trast, a study of high-skilled IT professionals by [Gibbs et al. \(2023\)](#) pointed to a productivity loss of 8 to 19%, which the authors attribute to increased coordination costs, in line with concerns that remote work impedes collaboration and can skew promotions towards individual metrics ([Cappelli and Nehmeh, 2025](#)). The emerging, though not definitive, consensus suggests that hybrid models may be productivity-neutral, while fully remote work could be associated with a productivity decline of around 10% ([Barrero et al., 2023](#); [Bloom et al., 2024](#)). The implications may also extend to future earnings: [Castex and Dechter \(2025\)](#) find that while current wages are unaffected, men with higher WFH hours seem to experience slower wage growth – a finding not mirrored for women, perhaps hinting at disparities in remote-based human capital accumulation.

This tension creates what [Cullen et al. \(2025\)](#) term the ‘WFH Wage Puzzle.’ Despite a high willingness-to-pay *for* WFH (a 25% wage sacrifice in their tech-worker sample), their work rejects a simple compensating differential, noting that remote jobs appear to pay slightly *more*. A potential resolution to this paradox is to separate the *treatment* effect (the amenity discount) from a *selection* effect (a sorting premium). WFH is not randomly assigned; it is more common among high-earner, high-education workers ([Pabilonia and Vernon, 2025](#)). Firms, particularly in competitive labor markets, may now be using WFH as a non-pecuniary tool to *attract* specific talent ([Ham et al., 2023](#)). This dynamic is further complicated by the geographic enlargement of the labor market. This expansion may enable “wage arbitrage,” with firms in high-cost-of-living (HCOL) areas hiring from low-cost-of-living (LCOL) regions ([Lightcast and Revelio Labs, 2023](#)). Concurrently, it may empower specialized LCOL workers by breaking their “local skill remoteness” and expanding their outside options, thereby increasing their bargaining power ([Macaluso, 2025](#)).

Finally, our paper is methodologically related to an emerging literature that uses large-scale text data from online job postings to measure labor market trends ([Hansen et al., 2023](#)). Specifically, we follow the approach of using a BERT-based model to classify job characteristics from unstructured text ([Cornejo et al., 2025](#)). By developing a sophisticated two-stage classification to distinguish between fully remote and hybrid roles, we provide a more nuanced measure of remote work availability and contribute to the toolkit for text-as-data analysis in labor economics.

3 Data

Our analysis is based on data from the Chilean online job board `trabajando.com` (TC). Our dataset captures comprehensive records of all job advertisements posted, all job seekers registered, and all job applications submitted between January 2008 and June 2024 (inclusive).

While firms pay to post vacancies, job seekers use the site for free.

Job ads contain rich information about the positions and the firms posting them, including specific hiring standards and compensation details. A distinctive and highly valuable feature is the requirement by the website for employers to enter the “*approximated net monthly wage*” (*salario líquido mensual aproximado*) in Chilean Pesos (CLP) for every advertised position. Employers must input a single monetary amount, not a range. Firms also choose whether to make this number visible to applicants or hide it. Ads with hidden wages are predominant, accounting for approximately 75% of all job ads on the platform. This access to wages for the large majority of postings circumvents a significant sample selection problem typically found in other online job board data (Batra et al., 2023). While one could argue that these *hidden* wages may not be informative (since most are never revealed to applicants), Banfi and Villena-Roldán (2019) show that they are in line with shown wages by comparable job ads, while Choi et al. (2025) show that wages in TC follow closely wages in survey data on newly hired workers in the Chilean economy.

Job ads specify prerequisites such as required level of experience (in years), required education (including college major), and indicators for specific skills (e.g., computing knowledge, foreign language, general/specific knowledge). Each job ad includes a job title, an occupational code,¹ a geographic location (region), and, crucially, the number of open positions (vacancies). The vacancy count enables us to weight job ads appropriately, ensuring estimates accurately reflect the actual flow of labor demand, especially since lower-paying jobs often advertise a higher number of positions.² In terms of firm details, we have information on size (number of employees in brackets) and a one-digit industry code (non-standard, created by TC). However, given the likely presence of recruiting firms in TC –which hire in behalf of some client firm and therefore mix the information of the hiring firm (in the ad) and the posting firm (in the firm directory)–, we follow Banfi and Villena-Roldán (2019) by flagging firms that in any quarter post either more vacancies than half their alleged firm size or those who post more than the top 1 percent of posters in that quarter.

Along these quantifiable variables, each job ad presents further information in the form of unstructured text in the job description. We use this corpus of text for the constructions of two extra variables: whether a job can be done remotely (WFH) and whether job entails full vs. part time.

We complement this expansive amount of information on job positions with information

¹These encompass 160, 5-digit ISCO08 labels provided by SABE (*Sistema de Análisis de Bolsas de Empleo*). We use the updated version launched in July 2024. SABE is funded by OTIC SOFOFA and developed by the Web Intelligence Centre, Universidad de Chile, in strategic alliance with the Servicio de Capacitación y Empleo (SENCE). See Cornejo et al. (2025).

²See Choi et al. (2025).

on the individuals who apply to them. For job seekers, we have information on date of birth, gender, nationality, marital status, region of residence, educational attainment, and self-reported years of experience. Applicants also report their *expected salary* (*expectativa de renta*), a net monthly figure they anticipate receiving after the job search process. As with posting firms, this expectation is observable to us, even if the applicant chooses to conceal it from potential employers. Additionally, we observe their current labor status (employed or non-employed).

Job seekers and job ads are linked through an application database, where we observe all applications from all job seekers to each job ad in our sample. Using this, we can construct two useful pieces of information for individual ads: average characteristics of individuals applying (along different dimensions, such as age, gender, location, etc.) and, information on *competing* jobs, i.e., the characteristics of positions where applicants have applied to *besides* ones own application.

4 Methodology: Identifying WFH and Work-Hour Arrangements

Our core methodological approach is a multi-stage classification process designed to overcome two measurement challenges in job ads. The first is to identify the specific mode of remote work. The second is to ensure the comparability of work hours by consistently identifying full-time job ads. This problem arises from a change in the data at the platform-level. Beginning in June 2020, *trabajando.com* introduced a “remote” category label as an option for their categorical variable describing work modality of the job. This was introduced in addition to the already existing categories of “full-time”, “part-time”, and “other”. Although intended to highlight WFH jobs, this new flag impedes learning whether WFH-labeled job ads were full-time or part-time.

Our analysis of wage rates requires comparing jobs with homogeneous work hours. In our empirical analysis below, we focus on full-time positions. This is justified by the institutional context of the Chilean labor market, which has a strong focus on the 45-hour workweek for full-time contracts (beyond 45 hours, overtime rates increase by 50%). Indeed, data from national surveys show that 60% or more of full-time private sector employees report a typical 45-hour workweek, a pattern that is applicable in most industries (Choi et al., 2025).

To solve the missing data problem, we classify both the WFH modality and the full-time status using the text in the job postings. This is accomplished by fine-tuning Large Language Models (LLMs) on a high-quality, manually labeled training dataset created by a team of

human annotators.

4.1 Construction of the classification dataset

Our database of job ads spans from 2008 to 2024. This 16-year period includes major structural breaks, such as the 2019 “social unrest” in Chile and the COVID-19 pandemic, which radically altered WFH preferences. To ensure that our training data are representative of these shifts, we partition the sample into four time periods.

1. Baseline: 01/01/2008 - 30/09/2019
2. Social Unrest: 01/10/2019 - 31/03/2020
3. Pandemic onset: 01/04/2020 - 31/12/2020
4. Pre-pandemic normalization: 01/01/2021 - 31/04/2024

To build a balanced training dataset, we first pre-identify potential WFH ads using an extensive dictionary of 1,835 keywords, phrases, and N-grams in Spanish and English. We augmented this list with keywords from the OECD and [Draca et al. \(2022\)](#) work from home keys, creating a final dictionary of 2,097 keywords. Following [Aksoy et al. \(2022\)](#), we classify these into four “buckets”:

- Specific WFH phrases
 - Bucket 1: Negated specific WFH phrase
 - Bucket 2: Non-negated specific WFH phrases
- Nonspecific WFH phrase
 - Bucket 3: Generic work from home phrases (e.g., 'tele', 'remoto')
 - Bucket 4: Non-generic WFH phrases (e.g., 'virtual', 'trabajo inteligente')

Generic WFH phrases are those that include a broad phrase or even a part of words that are necessary to communicate work from home. For example, the word 'tele' that by itself is also a subword that could imply work from home, for example in 'teletrabajo', 'telemarketing' or 'teleoffice', other Spanish words like 'remoto' or 'distancia' imply work from home in phrases like 'trabajo remoto', 'trabajo en forma remota' or 'trabajo a distancia'.

Non-generic WFH phrases are not specific or generic and are not necessarily related to work from home. For example, the words 'virtual' or 'inteligente' in Spanish are used as euphemisms to imply modern work from home or hybrid schedules in phrases like 'horario

virtual’ or ‘trabajo inteligente’. Finally, the negated specific phrases are built from the specific ones by using common negation words in Spanish like ‘no’ and ‘ni’.

We use this dictionary, along with an original WFH flag in the database for ‘teletrabajo’ to identify potential WFH job ads. This pool forms the basis for our manual classification sample. Figure 1 shows the distribution of ads identified by this dictionary approach across our time partitions, a reference for the keywords used can be found on appendix A.1.

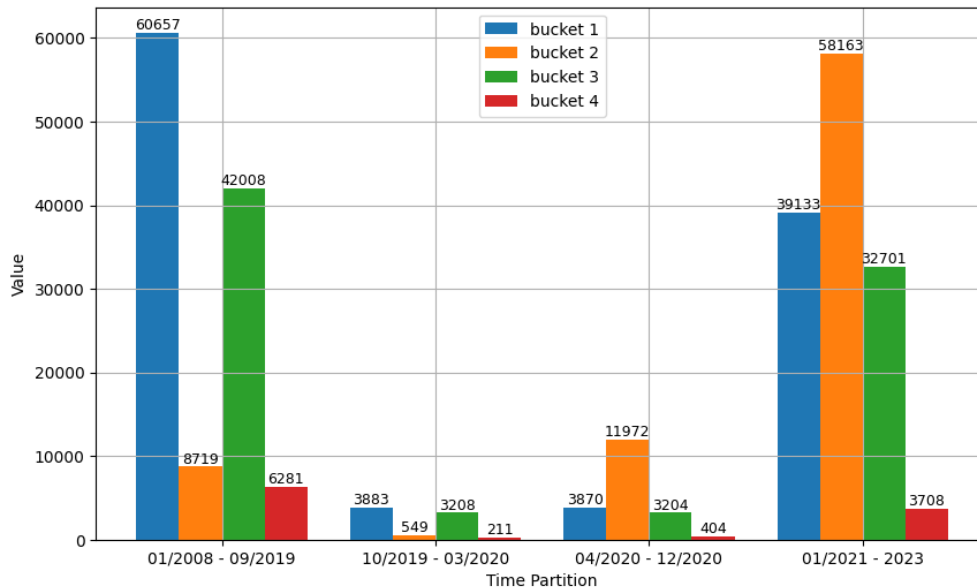


Figure 1: WFH counts by time partition and bucket (dictionary approach)

4.2 Creating a Random Sample

To ensure the diversity of our training sample and avoid overestimating WFH prevalence from recycled ads, we implement a de-duplication strategy. We first identify exact duplicates within six-month windows. We then use a TF-IDF (Term Frequency-Inverse Document Frequency) vectorization algorithm³ to identify “near-duplicates,” defined as ads with a cosine similarity of 0.85 or higher over their vectorized counterparts.

After removing these duplicates and near duplicates, we draw a final stratified random sample of 8,000 job advertisements. The sample is stratified equally across our four time partitions and four buckets, ensuring that each of the 16 strata is represented. This guarantees a robust representation of all WFH language types, particularly those concentrated in the post-COVID periods.

³This algorithm measure the importance of a specific word within a context adjusted by the frequency of words in the given context, is commonly used in natural language processing for identifying synonyms

4.3 A Two-Task Manual Classification Process

We recruited annotators by postings on social media (LinkedIn) and trained a team of 50 classifiers from a pool of 123 university students and recent graduates in Chile, selecting them based on performance on a standardized classification test of 13 manually pre-selected advertisements with different levels of 'difficulty' defined as the direct mention of work from home or not. Then these annotators were assigned the 8,000-ad sample and tasked with performing two separate classifications for each ad, using a custom-built web application ⁴. Each ad was classified by two different annotators to measure inter-annotator agreement.

Task 1: WFH Modality Classification. The first task involved a two-round process to determine the ad's WFH status.

- **Round 1 (Binary WFH):** Taggers first made a simple binary choice: does the ad mention remote work (WFH) or not (Not-WFH)? As shown in Table 1, the agreement rates were high (more than 90%). Figure 2 shows the individual disagreement rates, which we used to filter out five low-performing taggers.

Category	Number of tags	Percentage of tags ⁵	Percentage of agreement
Not WFH	11120	69.5%	95.65%
WFH	4880	30.5%	90.25%
Total	16000	100%	

Table 1: First Round (WFH Modality) classification results.

- **Round 2 (Granular WFH):** Ads identified as WFH in Round 1 were passed to a second round. The 45 best-performing taggers classified them into one of six granular categories: Fully Remote, Temporary Remote, Semi-remote (1-2 days on-site), Semi-on-site (1-2 days remote), Unclear Remote, and Not Remote (to catch inconsistencies). The results in Table 2 show, as expected, lower agreement rates for these more nuanced distinctions.

Task 2: Work-Hour Arrangement Classification. To overcome the platform's data limitation, annotators performed a second, simultaneous classification task during the first round. They were asked to read the full ad text, title, and requirements to determine if the position was Full Time, Partial (e.g., part-time, internship, freelance) or Not clear. The results for this task, shown in Table 3, demonstrate a very high level of inter-annotator agreement, confirming that full-time status is clearly discernible from the ad text.

⁴The classification tool is a simple javascript-python application with a database attached and real time user session logging that saves progress and assigns each ad to two different classifiers at random

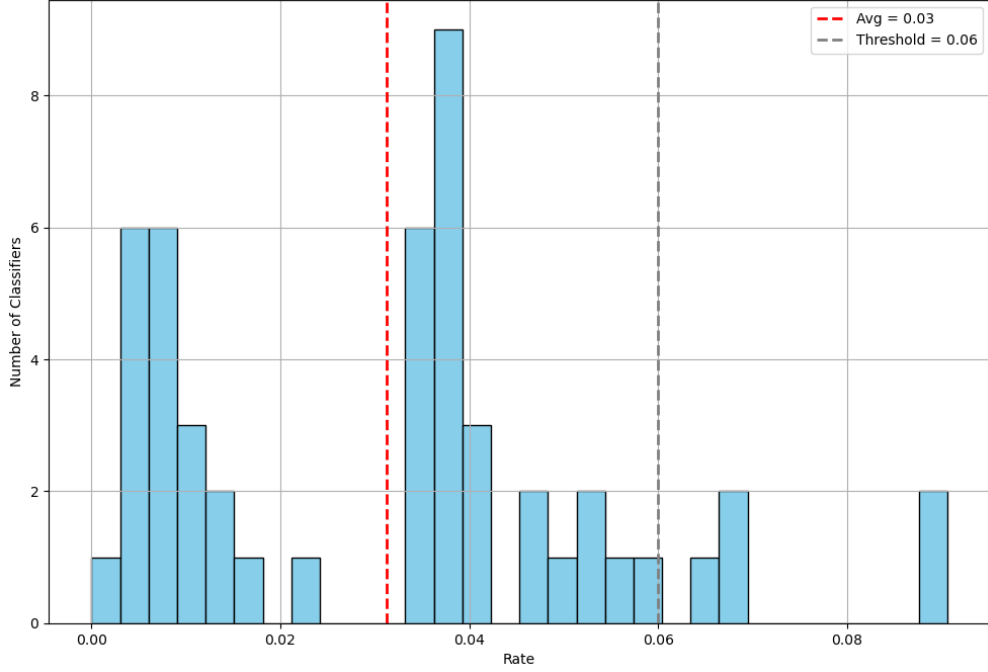


Figure 2: Disagreement rate by classifier, first round (WFH Modality)

Category	Tags (N)	Tags (%)	Agreement (%)
Fully remote work	1546	30.14%	73.31%
Temporary remote work	527	10.27%	61.65%
Remote work with onsite schedule	807	15.73%	40.10%
Partially on site with remote schedule	579	11.27%	38.52%
Unclear remote work	823	16.04%	31.89%
Not remote work	848	16.53%	72%
Total	5130	100%	

Table 2: Second Round (WFH Modality) classification results.

Category	Tags (N)	Tags (%)	Agreement (%)
Full-Time	10454	65.3%	92.2%
Part-time	2056	12.9%	82.4%
Not clear	3490	21.8%	75.4%
Total	16000	100%	

Table 3: Work-Hour Arrangement classification results.

4.4 Refining Categories and Ensuring Data Quality

For the WFH modality task, the second round revealed significant ambiguity, particularly for hybrid categories. We therefore consolidated these labels into a more robust three-way classification: WFH (Fully Remote), Hybrid (comprising Temporary, Semi-remote, Semi-on-site, and Unclear) and Not-WFH. Table 4 shows that this aggregation significantly improved the agreement rates.

Category	Number of tags	Percentage of tags	Percentage of agreement
WFH partial (hybrid)	2736	53.33%	79.29%
WFH	1546	30.14%	73.32%
Not WFH	848	16.53%	72%
Total	5130	100%	

Table 4: Second Round (WFH Modality) classification results with collapsed hybrid categories.

4.5 LLM Training Datasets

Disagreement between annotators can result from genuine ambiguity in the ad text or from a classifier error. One potential indicator of the quality of the annotation is the time devoted to do it, discarding recording times longer than 5 minutes because they likely imply a rest pause in the process. We analyzed the annotator speed (Figures 3 and 4) but found no strong correlation with disagreement rates.

We therefore solely rely on inter-annotator agreement (Table 5) to define three final training datasets for both classification tasks (WFH Modality and Full-Time):

- **Conservative database:** Keeps only ads with perfect agreement across all taggers and rounds. In this database we are removing everything that could be considered ambiguous, including ads that could be inherently unclear.
- **Naive database:** Keeps all classifications from all taggers, retaining all disagreement.
- **Quality Control database:** Our preferred set of consistent annotators. We remove from the first round those taggers showing disagreement rates higher than 6% and from the second round taggers with rates higher than 20%. This choice thus retaining realistic ambiguity while removing clear error based on the average agreement for each round.

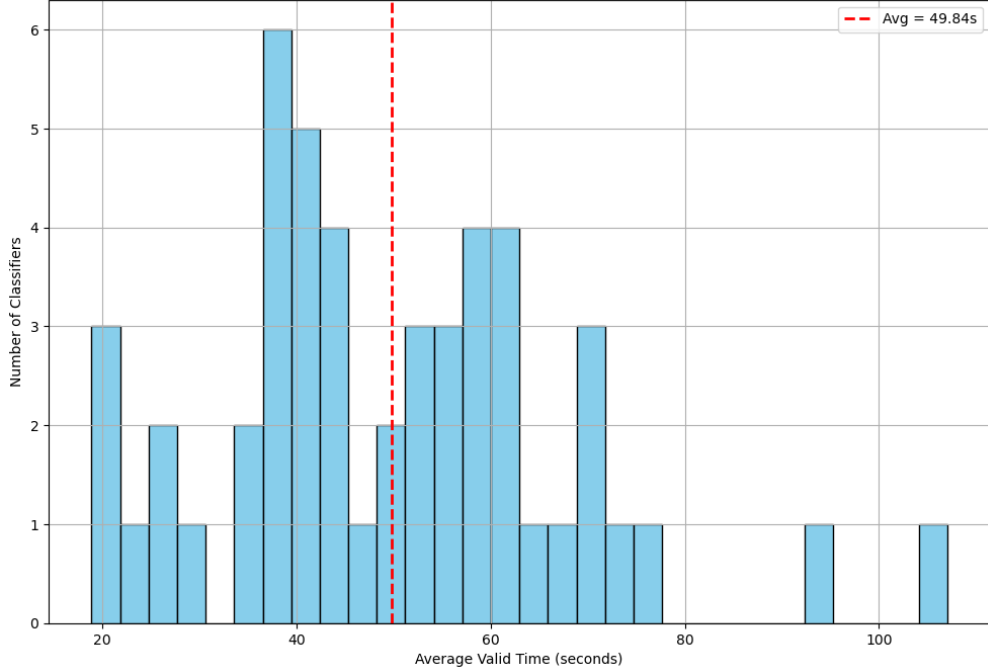


Figure 3: Average Valid Time per Classification by Classifier (seconds), first round

Round 1, Tag 1	Round 1, Tag 2	Round 2, Tag 1	Round 2, Tag 2	Count
WFH	WFH	WFH	WFH	643
WFH	WFH	Hybrid	Hybrid	1163
Not WFH	Not WFH			5568
...	...	...	...	...

Table 5: Possible combination of classification outcomes for WFH Modality task.

4.6 Models and Tag Generation

Now we used this different databases to fine-tune text classification models for work from home modality and full time job status.

Model 1: WFH Modality. We fine-tuned three benchmark transformer models for Spanish text classification: DistilBERT (Sanh et al., 2019), RoBERTa (Liu et al., 2019), and BETO (Cañete et al., 2020). As shown in Table 6, the Spanish-specific BETO model achieved the highest accuracy (0.911) for the three-way WFH classification task, confirming its suitability.

Considering just the BETO model we can compare the nuances between the three different datasets and evaluate the effect of ambiguity in the predictions.

In table 7 we evaluate pairwise concordance among the same model trained on the different datasets using two complementary indices, the overall agreement rate that accounts

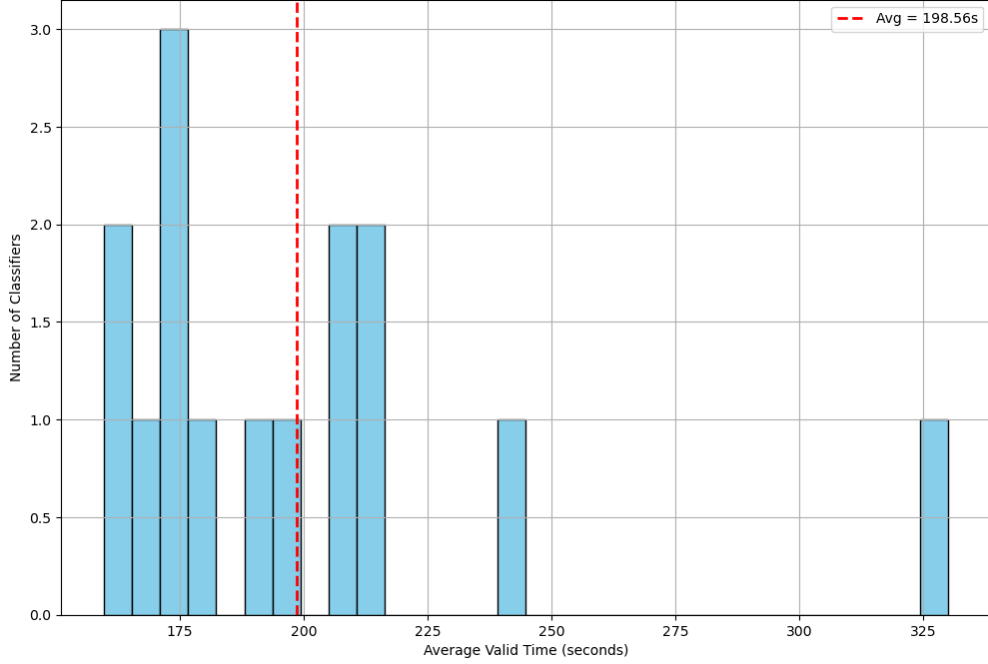


Figure 4: Average Valid Time per Classification by Classifier (seconds), second round

Table 6: Models performance for **WFH Modality** (*quality control dataset*). Best scores in **bold**.

Model	Acc.	F ₁	Prec.	Rec.	Runtime (s)	Samples/s
DistilBERT _M	0.878	0.870	0.868	0.878	9.47	147
RoBERTa _{Base}	0.883	0.883	0.883	0.883	1.80	773
RoBERTa _{Large}	0.899	0.899	0.899	0.899	5.19	268
BETO _{dccuchile}	0.911	0.912	0.913	0.911	1.91	730

for the proportion of instances for which two models yield identical class labels is closer to our previous inter agreement measure and it is an interpretable, accuracy-like measure of concordant decisions, but it is known to be inflated when marginal classes are extreme, as in heavily imbalanced datasets like ours where we know that not WFH tags are far more frequent.

Cohen’s κ tries to correct this to ensure better reliability

$$\kappa = \frac{p_0 - p_e}{1 - p_e} \quad (1)$$

where p_e is the expected agreement under independence induced by the models marginal label distributions. By adjusting for p_e , κ provides a more conservative assessment of reliability when the positive class dominates as is in our problem where not work from home tags are more frequent.

Table 7: Agreement metric for training datasets BETO Model

Comparison	Agreement rate p_0	Cohen’s κ
Naive vs Quality control	0.986	0.856
Naive vs Conservative	0.993	0.924
Quality control vs Conservative	0.985	0.847

We can see that the class prevalence of the positive label (no WFH) is very high for all three models, indicating pronounced imbalances. This raises the possibility of optimistic agreement rates and justifies the emphasis on κ that tries to account for this.

Considering everything for the overwhelming majority of cases the models arrive at the same decision. κ coefficient despite the positive class prevalence inflation of p_0 , remains in a strong range, we can say that κ differentials across pairs meaningfully capture subtle divergences that overall agreement obscures.

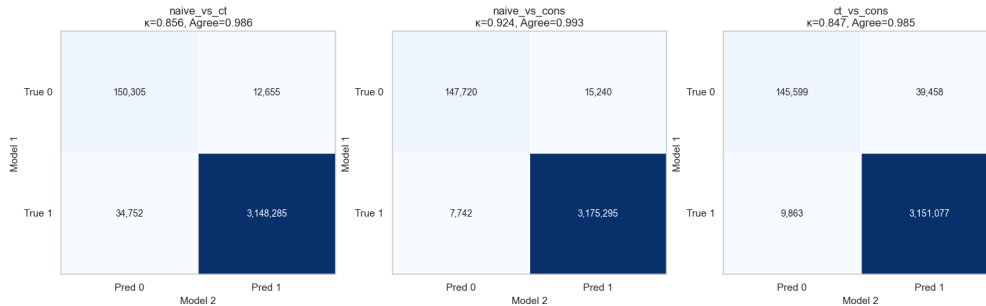


Figure 5: Confusion matrix comparisons

We also report cross tabulations 2×2 confusion matrices across models on figure 5 and each model’s positive-class prevalence to contextualize potential sources of agreement inflation and to diagnose directional asymmetries in disagreements.

The bulk of concordance resides in the (pred 1, true 1) cell, as expected from high positive prevalence this means most of not WFH model predictions that goes in line with most of job ads being not WFH in the whole database. The asymmetries between other cells reflect which model tends to be more ”liberal” (that means it trust the large category so it predicts not WFH more often) or ”conservative” (distrust the large category so it does not predicts not WFH often) at the decision boundary.

If we compare Naive vs Conservative datasets we see that this pair exhibits the strongest reliability and the fewest disagreements. Prevalences are very closely matched, suggesting near identical decision surfaces with only minor differences for a small fraction of borderline instances. This means that replacing one with the other would have negligible downstream effects for most applications.

Comparing Naive vs Quality control still have high agreement rates, yet κ is meaningfully lower than for Naive vs Conservative comparison. The Quality control model has lower positive prevalence, consistent with a slightly stricter boundary. The asymmetry in disagreements confirms that Quality control rejects a subset that Naive admits. These rejections likely correspond to low-margin or noisier cases that the Quality control training dataset is considering differently. Thus, Quality control introduces a modest conservatism relative to Naive without wholesale reconfiguration of the hypothesis class.

This is complemented with the Quality control vs Conservative models comparison. This is the weakest pair though still strong. Conservative has the highest positive prevalence, and disagreements goes toward Conservative predicting not WFH where Quality control predicts WFH, here we see that the general overrepresented category of not WFH works as a "secure choice" for the models in general and more for the conservative model that don't assume the "risk" of giving it another label, acting as the default label, we can safely assume that this behavior could also appeared among the taggers, due that in the tagging exercise it was relatively easy to discern that most of the tags are not WFH with a given strike while classifying.

The pattern indicates that, relative to Quality control, the Conservative model is not "conservative" in the sense of predicting fewer positives; rather, "Conservative" here reflects a training-data variant rather than a stricter decision policy. Empirically, it is the Quality control model that functions as the more selective classifier given that default behavior of the Conservative approach and the overconfidence of the naive model. The magnitude of κ suggests that Quality control adjustments affect a non trivial subset of cases at the boundary that could be difficult to assign and inherently ambiguous, even incorporating this nuances in the Quality control approach still give a high global alignment.

From this comparison of the models we can assume robust consistency of learned decision functions across the three training regimes. This is reassuring given the scale and skew of the data, the modeling pipeline is not highly sensitive to dataset curation choices for the vast majority of instances.

We have also detected boundary advertisements where the distinction is subtle, here is where the differences between the models concentrate, near the decision threshold. The lower positive prevalence and asymmetry of the Quality Control model of disagreements against Naive and Conservative suggest that hard consistency could mislabel a thin slice of ambiguous examples. So to safely include this real ambiguity and at the same time safely avoid mislabelling in these edge cases, the Quality control model could yield meaningfully different error tradeoffs and is nearly identical at the aggregate level.

Even so, if further applications want to force the model to choose an option based on

the context (strong not WFH dominance in conservative approach) or allow more ambiguity in edge cases (naive approach), both Naive and Conservative are near-substitutable in downstream deployments. Quality control provides a slightly more selective alternative with modest divergence. So, the Naive and Conservative models are functionally interchangeable for most applications, whereas the Quality control model is modestly more selective, disagreeing primarily by reclassifying a small set of borderline positives as negatives. These differences are statistically meaningful, yet operationally localized to the decision boundary.

Model 2: Full-Time Status. We followed an identical procedure to train a classifier for the binary work-hour arrangement (Full-Time vs. Other). Given its superior performance, we again used the BETO model fine-tuned on our *Quality Control dataset*. The results, presented in Table 8, show that the model achieves a very high accuracy of 0.927, confirming our ability to reliably identify full-time jobs from ad text.

Table 8: Models performance for **Full-Time Status** classification (*quality control dataset*).

Model	Acc.	F ₁	Prec.	Rec.	Runtime (s)	Samples/s
BETO _{decuchile}	0.927	0.915	0.917	0.921	1.50	781

This two-model pipeline is the foundation of our empirical strategy. It allows us to first use the Full-Time classifier to identify the correct universe of ads (over two million) for our analysis and subsequently apply the WFH-Modality classifier to this subset. This process ensures that we are comparing monthly wages for comparable (i.e., full-time) jobs, thereby isolating the effect of remote work from the confounding factor of work hours.

5 Econometric Framework

To estimate the causal impact of remote work probabilities on wages, modality offerings, and job competition, we need to consider strategic interactions between employers.

5.1 Strategic Interactions

We characterize the labor market as an oligopsonistic environment in which firm decisions about wages and work modalities are not made in isolation but as strategic reactions to the behavior of their competitors. This framework follows the logic of social effect models in which strategic interactions in networks are rationalized as games in which firms with interdependent objective functions respond to the average strategies of other players as in [Bramoullé et al. \(2014\)](#) and [Acemoglu et al. \(2015\)](#). In our context, following [Banfi](#)

et al. (2025), we leverage the bipartite network of job ads and applicants linked through applications to identify direct competitors, defined as ads that receive applications from the same pool of workers.

If we take a reference ad a we denote the set of applicants or workers applying to it as $\mathcal{W}_a$. Similarly, the set of ads that receive applications from a reference applicant w is denoted as $\mathcal{A}_w$. A natural set of competitors for a is the set of ads that receive applications from applicants to a . This would be equivalent to $\mathbb{A}(\mathcal{W}_a) \equiv \cup_{w \in \mathcal{W}_a} \mathcal{A}_w$. If ads competing with a increase their wage or change their job modality to remote (WFH) or hybrid (HYB), these actions likely affect a 's optimal strategy. In particular, if the objective of firms is quadratic, their optimal (interior) response function in a Nash equilibrium is linear, such that each firm behaves considering the average behavior of their competitors.

Following these insights, we model the log wage ($\log w_a$) and the decision to offer remote work (WFH or Hybrid) for job ad a as simultaneous decisions determined by the job's characteristics and the average behavior of competing firms. The structural equation for wages is defined as:

$$\log w_a = \beta_1 \text{WFH}_a + \beta_2 \text{HYB}_a + X_a \beta + \delta E[\log w_{-a} | X_a] + \epsilon_a \quad (2)$$

where WFH and HYB represent the predicted probabilities (by the classification process described above) for ad a and X_a is a vector of observed job characteristics. The term $E[\log w_{-a} | X_a]$ captures the average wage offered by the competitors of a . The parameter δ measures the strength of strategic interactions; if $\delta = 0$, the market is Walrasian: ads are price-takers, and wages simply reflect the pricing of job attributes. If $\delta > 0$ ($\delta < 0$), wage posting decisions are strategic complements (substitutes). Crucially, the modality variables of D1 competitors are not included as endogenous regressors in the equation (2). We assume that the effect of cross-competition on remote modalities is of secondary importance to affect wages, as the reference ad can adjust its own wage to compete effectively.

Simultaneously, the decision to offer a specific modality is also subject to strategic interactions:

$$\text{WFH}_a = \alpha_1 \log w_a + X_a \alpha + \lambda_1 E[\text{WFH}_{-a} | X_a] + \lambda_2 E[\text{HYB}_{-a} | X_a] + v_a \quad (3)$$

$$\text{HYB}_a = \gamma_1 \log w_a + X_a \gamma + \phi_1 E[\text{WFH}_{-a} | X_a] + \phi_2 E[\text{HYB}_{-a} | X_a] + \varsigma_a \quad (4)$$

Here, λ and ϕ capture how a firm's modality decision is influenced by the prevalence of remote work among its competitors. The underlying logic is that the modalities offered by D1 competitors influence the choice of modality of the reference ad a . In these equations, the focal ad's own log wage is included, while the average log wage of D1 competitors is

excluded to focus on the effect of the firm’s own wage offer on its modality decision.

Other outcomes Y can be a function of wages and modalities and can also be endogenous, although we do not stress the competition along those dimensions

$$Y_a = \gamma_1 \text{WFH}_a + \gamma_2 \text{HYB}_a + \gamma_3 \log w_a + X_a \gamma + \xi_a \quad (5)$$

5.2 Identification Strategy: The Application Network

Estimating these equations presents a “reflection problem” (Manski, 1993). A job ad a makes strategic decisions, in this case the offered wage and the modality, taking into account the strategic reactions of its competitors. In turn, the strategic choices of competitors, in response to job a ’s move, affect job ad a back. Therefore, the average behavior of other competitors becomes an endogenous regressor.

To solve the reflection problem, we use the bipartite network structure of job applications to identify ad-specific local labor markets and valid instruments, mimicking the strategy in Bramoullé et al. (2009) of “friends of friends who are not my friends”. Applied to our context, for every reference ad a , the set of its competitors $\mathbb{A}(\mathcal{W}_a)$ faces, in turn, competition from other competitors in the set $\mathbb{A}(\mathbb{A}(\mathcal{W}_a))$ which do not include the reference ad a . Therefore, when firms in set $\mathbb{A}(\mathbb{A}(\mathcal{W}_a))$ change their behavior, they induce a change in the behavior of a , but the reaction of a does not affect the behavior of posters in $\mathbb{A}(\mathbb{A}(\mathcal{W}_a))$.⁶ Note that these sets can be thought of competing ads with certain *degrees* of separation: $\mathbb{A}(\mathcal{W}_a)$ represents competing ads one degree of separation (D1) away from a ; $\mathbb{A}(\mathbb{A}(\mathcal{W}_a))$, represents two degrees of separation (D2).

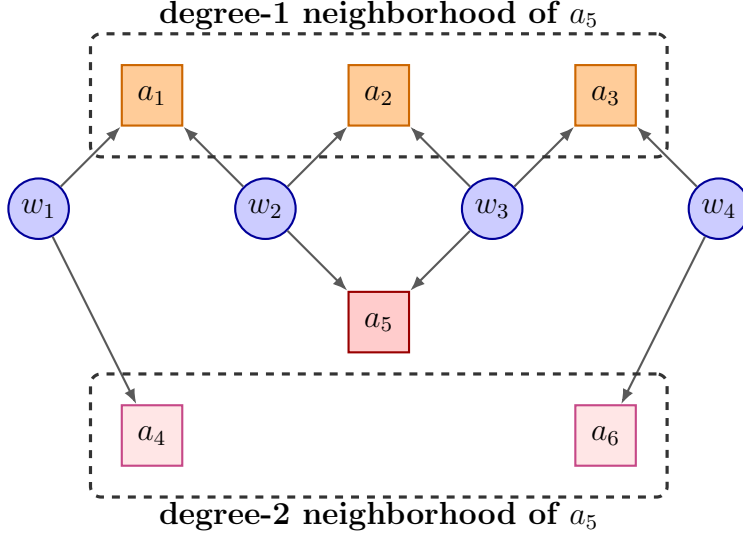
Figure 6 illustrates a simple example in which the ads are represented by squares and the applicants by circles. Arrows represent applications that create links on the network. Taking the reference ad a_5 , its 1-degree competitors are the ads that receive applications from $\mathcal{W}_5 = \{w_2, w_3\}$, i.e., $\mathbb{A}(\mathcal{W}_5) = \{a_1, a_2, a_3\}$. In turn, competitors of competitors of a_5 are ads linked to $\mathbb{A}(\mathcal{W}_5)$ that exclude those in this set, that is, $\mathbb{A}(\mathbb{A}(\mathcal{W}_5)) - \mathbb{A}(\mathcal{W}_5) = \{a_4, a_6\}$.

To implement the aforementioned strategy, we need to compute the average responses of the relevant set of competitors, either D1, as an endogenous regressor, or D2, as an exogenous source of variation, weighted by their correspondent vacancy count. Doing so in a consistent way requires computing the sums of the attributes of competitors for each ad a using the already introduced denotation. We define the network sums for a variable X at different degrees of separation:

- **Degree-0 (S^0):** The variable of the ad itself, $S^0(a, X) = X_a$.

⁶This rationale is similar to the circular city model of Salop (1979), in which firms compete only directly with adjacent competitors.

Figure 6: Application Network Diagram



- **Degree-1 (S^1):** The sum of variables X of all advertisements competing for the same pool of applicants.

$$S^1(a, X) = \sum_{b \in \mathbb{A}(\mathcal{W}_a)} X_b$$

- **Degree-2 (S^2):** The sum of characteristics of ads that share applicants with competitors of a .

$$S^2(a, X) = \sum_{b \in \mathbb{A}(\mathbb{A}(\mathcal{W}_a))} X_b$$

By definition, the sum $S^j(a, X)$ includes all the ads in $S^{j-1}(a, X)$. Hence, to compute the average of the “ring” surrounding $S^{j-1}(a, X)$, we need to subtract it from $S^j(a, X)$. In particular, a sum of X for the ring surrounding the reference ad a should be $S^1(a, X) - S^0(a, X)$. The sum of X for the ring surrounding the competitors of the ad a is $S^2(a, X) - S^1(a, X)$. With these definitions, we can define a weighted degree-1 network average of a variable X for an ad a as

$$\overline{X}_{-a}^1 = \frac{S^1(a, vX) - S^0(a, vX)}{S^1(a, v) - S^0(a, v)}$$

Similarly, a degree-2 network average of a variable X for an ad a is defined as

$$\overline{X}_{-a}^2 = \frac{S^2(a, vX) - S^1(a, vX)}{S^2(a, v) - S^1(a, v)}$$

If the averages are vacancy-weighted, v_a is the number of vacancies posted for ad a . If the average is ad-weighted, then $v_a = 1$.

The identification strategy relies on a standard exclusion restriction in instrumental variable settings. In this case, the behaviors of competitors of competitors (degree-2 neighbors) affect the focal employer only through their impact on the direct competitors (degree-1 neighbors). Specifically, $E[\log w_{-n}|X_n]$, $E[\text{WFH}_{-n}|X_n]$, and $E[\text{HYB}_{-n}|X_n]$ in the structural equations are instrumented using these degree-2 network averages.

Given the IV strategy that we advocate here, it is important to revisit the specification of the wage equation 2. In addition to the argument that the strategy of the ad a to face their D1 competitors is of secondary importance when the wage offered setting is possible, we also notice that attempting to instrument these cross-competition variables would necessitate an additional instrument that could make substantially difficult to estimate the primary parameter of interest: how the offered wage reacts to an arguably exogenous variation in remote modalities.

We make a symmetric argument for the case of the modality equations (3) and (4). The only difference in these cases is that the modality choice is trinary, so two endogenous choices are needed to account for the competition effects in D1. An attempt to estimate the cross-competition effects of competitor wages on modalities would require the use of additional instruments to generate an independent exogenous source of variation for $E[\log w|X_{-a}]$ in addition to the one of D2-competitors for $\log w_a$. Again, the parameters of interest here are the reaction of modality choices of ad a when there is an exogenous variation of the wage offered for the same reference ad.

In equation (5) for other outcomes we do not emphasize the competition dimensions. In these cases, we only need to estimate the response of an outcome y to exogenous variations of the offered wage and modalities. We will use the average behavior of D2 competitors of the reference job ad in offered wages and modality choices to estimate these effects. A possible objection to the validity of this instrument is a potential connection among firms in another way, such as geography or occupation, classical variables used to define local labor markets (e.g. Şahin et al. (2014)). We address these concerns by controlling for occupation and firm-fixed effects in some specifications, as firms are attached to just one location. On a deeper level, rather than relying on ex ante determined and available characteristics, the network approach provides a definition of labor markets based on actual applications (Banfi et al., 2025) or effective job transitions Jarosch et al. (2024). If geography, occupation, organizational climate, or some other characteristic defines local labor markets, it will be reflected in the actual applications of individuals.

In all of the equations, we control for several job characteristics that are likely predetermined before deciding on specific offered wages and work modalities including industry, occupation, and job titles. In each specification, we also include fixed effects for year and

month-of-year to control for trends, cyclical variation, and seasonal effects.

We also use fixed effects for firms, which control for firm specific amenities that could be confounded with WFH or HYB jobs, such as corporate culture, organizational climate, etc. However, the sample shrinks because firms that have posted only one job ad within the period inevitably drop out of the sample. Therefore, firm fixed effects estimates are likely the golden standard, but the sample is biased against small firms.

We refrain from controlling for requirements that employers themselves endogenously set to fine-tune applicant quality, such as education, work experience, or specific and general knowledge. The core requirements that define the position are captured by occupation and job title categories. In contrast, education and experience requirements that differ within the same job title are adjusted in a countercyclical manner (Modestino et al., 2020; Choi et al., 2025), illustrating their endogenous nature.

In the next section, we focus our attention on full-time jobs posted on TC between 2010 and 2024. As described above, a variable created by the board identifies full-time jobs until June 2020. After this date, we define a job to be full-time if it is both defined as "full-time" or "remote" according to TC *and* if our own classification post June 2020 deems the ad to describe a full-time position. In terms of the span of the sample, while we use information from 2008 onward to classify ads, we estimate the main effects using a post 2010 sample, given that 2008 and 2009 coincide with a time period in which TC penetrated the Chilean market significantly, which may bias results in terms of representativeness.

6 Empirical Results

6.1 Wage Behavior

Table 9 presents the estimates for the wage equation. Column (1) shows OLS estimates with an array of fixed effects, trying to make job ads comparable across industries, occupations, and job titles. Column (2) adds firm fixed effects.⁷ In both specifications, the coefficients on WFH and HYB indicate a wage premium of about 3% to 4%. Once we instrument for the endogeneity of the work modality and network formation in Columns (3)–(6), the WFH premium increases substantially. Without firm fixed effects, the WFH premium is close to 16%. When we include firm fixed effects the estimated effect increases to 24% in column (6). In stark contrast, the effect of hybrid job probability is estimated around -3.6%. Since the WFH probability is a continuous variable, the correct interpretation of its associated

⁷This column excludes observations associated with firms that posted only a single ad (by construction) and those we deem as recruiting firms.

coefficient is the percentage change of the offered wage when the WFH probability increases marginally, keeping the HYB probability constant. Hence, the probability increase in WFH mirrors a probability reduction of WFH.⁸

The IV results suggest that WFH and HYB, although jointly studied in many settings, are distinct work modalities. Our findings suggest that when employers have to offer WFH by market pressure motive due to changes in the WFH posting behavior of competitors, they increase wages due to a loss of local monopsonic market power to larger, remote-accessible markets. In contrast, an increased frequency of HYB among competitors produces a small negative wage effect. This suggests that hybrid jobs are still constrained by geography so that employers retain local monopsony power and the amenity value of partial remote work predominates among applicants. These results are aligned with findings related to the geography of workers (Akan et al., 2025).

The estimated coefficients for the wage posted by competitors imply a very strong evidence of strategic interactions (although when instrumented, the magnitude of the interaction decreases) since firms strongly align their wages with direct competitors.

Our IV estimates reject weak instruments as shown by the high Cragg and Donald (1993) (CD) tests. The usual rule-of-thumb critical value for a single endogenous regressor and a relative bias of 10% is 10, and for two endogenous regressors it is around 13.4 (Stock and Yogo, 2005). Our instruments generate a CD test several orders of magnitude above those values. Despite the fact that we did not find tabulated critical values for cases with three endogenous regressors in columns (5) and (6), the reported CD test values are far beyond the usual range. Therefore, the first stages are unusually strong.

Robustness: We consider whether these results hold during the onset and aftermath of COVID given the alleged structural change in the market. Part of this issue is already controlled in Table 9 since we control for year and monthly seasonal fixed effects. In the appendix A.4.1 we estimate the model for 2020-2024 in Table A3. Despite the massive structural transformation of remote modalities, several qualitative results are similar. The OLS estimation for the WFH and HYB jobs yields slightly higher results, and the IV estimates show somewhat lower but still higher returns. Since the IV results for the probability of HYB are now somewhat negative, around 5-11% wage penalty, we interpret that during and after the pandemic the amenity of full or partially remote work became more valuable for applicants who demanded lower wages for otherwise similar jobs. Other estimates of the strategic reaction of employers to the wages of their competitors remained quite similar to

⁸To approximate the effect of a larger shift from 0 to 1 WFH probability, we need to consider the initial HYB probability, since such a large change necessarily implies a reduction of HYB and on-site probabilities at once.

Table 9: Estimation of log wage equation (2) 2010-2024: OLS vs IV estimates

	(1)	(2)	(3)	(4)	(5)	(6)
WFH prob	0.046*** (0.0032)	0.034*** (0.0039)	0.17*** (0.0095)	0.28*** (0.015)	0.15*** (0.0097)	0.24*** (0.016)
HYB prob	0.033*** (0.0011)	0.033*** (0.0013)	-0.017*** (0.0035)	-0.019*** (0.0048)	-0.036*** (0.0036)	-0.036*** (0.0048)
D1-avg log wage	0.75*** (0.00075)	0.63*** (0.00091)	0.75*** (0.00075)	0.63*** (0.00091)	0.54*** (0.0085)	0.44*** (0.011)
Vac Obs	4564115	3344093	4558817	3340526	4558817	3340526
R2	0.67	0.75	0.18	0.13	0.17	0.12
Adjusted R2	0.67	0.74	0.18	0.12	0.17	0.11
CD test			238648.8	110626.6	12002.7	7917.3
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

the extended 2010-2024 period.

6.2 Modality Offers

Table 10 shows the OLS and IV estimates for equation 3, i.e., the probability of a reference job ad to offer WFH. Although both OLS and IV results find a positive impact of wages on the probability of posting a WFH ad, the effect is small, which shows that the WFH decision is mostly detached from wage offers. Within the IV results, the consideration of firm fixed effects and its consequential sample reduction makes a difference, especially when the three endogenous variables are instrumented.

The impact of competing ads that post WFH ads in our network definition is high and significant, and the effect is substantially stronger in IV estimates without firm fixed effects (Columns (3) and (5)). Considering column (6) as a benchmark, an increase of 1% of average probability of posting a WFH ad among competitors increases the probability of WFH posting by 1.16%.⁹ In contrast, the response of WFH postings to the probability of HYB modality among competitors is much smaller, and even mildly negative if we consider the IV estimates. This evidence points to the consideration of the WFH and HYB modalities

⁹Since these estimates are above 1, a stability concern of the expectations model could be raised. Nevertheless, the effect is essentially a local linear approximation. As noticed before, a large change of probabilities overall would imply a reduction of HYB probabilities and that some important nonlinearities kick in, something we are not considering here.

as quite different employer decisions, often competing against each other.

Turning to the results on the HYB posting decisions in Table 11 we find that the impact of wages on the probability of posting HYB is negative when we use the instruments for wage and D1 competition in WFH and HYB in columns (5) and (6). These estimates are interpreted as an increase of 1% in the offered wages due to higher competing wages induced by D2-competitors imply a drop of 0.12 - 0.069 percentage points in the probability of HYB. If employers are induced by their competing employers to offer higher wages, they turn to offer HYB less likely, suggesting that providing this amenity could be costly for them.

The response of HYB probabilities to D1 competition of WFH and HYB arrangements are consistent with strong segmentation between modalities in the sense that the network effect is specific: a higher prevalence of HYB among competitors strongly increases a employer's probability of offering HYB, but has a negative spillover on the probability of offering fully remote work. With milder effects of the negative cross-effect, the reverse is true for the WFH decision. This suggests that WFH and Hybrid arrangements function as distinct strategic choices rather than close substitutes.

Robustness: In the appendices A.4.2 and A.4.3 we re-estimate tables 10 and 11 for the sample 2020-2024 in tables A5 and A7, respectively. Again, whether the reaction of the job modalities and wages changed during and after COVID is a major concern.

The sensitivity of the WFH probability to offered wages remains low in the 2020-2024 sample, although a larger negative impact occurs in the case without firm fixed effects. Exogenously induced variations in wages do not sizably affect WFH posting. The sensitivity to WFH competition is somewhat smaller in this case, but is still a major determinant of WFH. The small negative effect of competitors' HYB posts still dissuades WFH posts. Turning to HYB probability estimation for 2020-2024, we find that a competitor-induced variation of wages affects HYB posting more importantly than in the whole sample. As in the WFH probability case, the interaction effects are somewhat milder but very significant, showing that key competition mechanisms work during hectic pandemic times, too.

6.3 Competition and Applicants' characteristics

In this section, we try to dig deeper into some mechanisms to understand the previous results.

Applications: We estimate equation (5) using log applications received by each ad as a dependent variable, using as a potentially endogenous regressors log offered wage, WFH and HYB probabilities, in addition to other controls in the form of fixed effects of year, seasonal month, industry, occupation, job title words, and eventually firm (with caveats on the resulting sample we discussed above).

Table 10: Estimation of WFH prob equation (3) 2010-2024: OLS vs IV estimates

	(1)	(2)	(3)	(4)	(5)	(6)
log wage	0.00088*** (0.000061)	0.00052*** (0.000072)	0.00025*** (0.000065)	0.00019** (0.000075)	-0.012*** (0.0024)	-0.00098 (0.0035)
D1-avg WFH prob	0.71*** (0.0013)	0.42*** (0.0015)	1.75*** (0.0026)	1.16*** (0.0026)	1.76*** (0.0026)	1.16*** (0.0027)
D1-avg HYB prob	0.022*** (0.00034)	0.029*** (0.00040)	-0.053*** (0.00073)	-0.030*** (0.00086)	-0.054*** (0.00077)	-0.030*** (0.00090)
Vac Obs	4564115	3344093	4558817	3340526	4558817	3340526
R2	0.11	0.21	-0.066	-0.042	-0.074	-0.042
Adjusted R2	0.11	0.21	-0.066	-0.053	-0.074	-0.053
CD test			731151.1	485425.4	1147.1	494.9
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

An increase in WFH and HYB probabilities generates moderate percentage increase in applicants according to OLS estimates, but much larger for IV estimates, particularly if firm fixed effects are included. Taking column (6) as reference, a 1 percentage point increase in the WFH probability increases the applications received by nearly 2%. Although highly significant, the estimated effects for the WFH probability are substantially milder but clearly positive.

The effect of log wages on applications depends largely on whether this variable is instrumented using D2-competitor averages. When not instrumented, an increase in log wages is positively associated with an increase in applications by approximately 0.04-0.11%, in line with the results in [Banfi and Villena-Roldán \(2019\)](#), suggesting that higher wages and applications are partially correlated. In contrast, when this variable is instrumented, the effect becomes large and negative. This result can be reconciled with the “portfolio effect” described in [Galenianos and Kircher \(2009\)](#) (and to some extent in [Acemoglu and Shimer \(1999\)](#) and [Albrecht et al. \(2006\)](#)): an increase in the wage of the reference job induced by higher average wages of its D2-competitors attracts more applicants, increasing the competition among them. Provided applicants are risk-averse, after a wage rise, they tend to *additionally* include jobs with lower wages and less competition in their application portfolios. In light of this theoretical framework, an exogenous positive wage shock in the D2-set of competitors induces a reduction of wages among the nearest competitors of the reference ad (D1-competitors). Then, the reference ad increases its wage to receive more applications

Table 11: Estimation of HYB prob equation (4) 2010-2024: OLS vs IV estimates

	(1)	(2)	(3)	(4)	(5)	(6)
log wage	0.0033*** (0.00016)	0.0037*** (0.00020)	0.0038*** (0.00017)	0.0030*** (0.00021)	-0.12*** (0.0063)	-0.069*** (0.0098)
D1-avg WFH prob	0.10*** (0.0034)	0.053*** (0.0043)	-0.20*** (0.0066)	-0.22*** (0.0072)	-0.18*** (0.0070)	-0.21*** (0.0076)
D1-avg HYB prob	0.95*** (0.00090)	0.89*** (0.0011)	1.33*** (0.0019)	1.23*** (0.0024)	1.32*** (0.0020)	1.22*** (0.0025)
Vac Obs	4564115	3344093	4558817	3340526	4558817	3340526
R2	0.33	0.49	0.18	0.15	0.085	0.12
Adjusted R2	0.33	0.48	0.18	0.14	0.085	0.11
CD test			731151.1	485425.4	1147.1	494.9
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

by providing a high-return but high-risk job (from an applicant’s perspective). Though not considered in the theoretical papers cited, in overlapped local labor markets such as the one portrayed in our data, the effect of the reference ad wage also induces changes in the composition of applications received. Therefore, the drop in applications due to an exogenous wage increase may also imply a higher quality of applicants, reinforcing the portfolio effect.

Labor market size: One hypothesis to explain the increase in wages under remote modalities, especially WFH, is that a higher frequency of WFH in competing employers reduces the monopsonic power of the firm. The geographical location of a job significantly determines the search behavior of the individuals applying to it and is a well established fact (Manning and Petrongolo, 2017; Marinescu and Rathelot, 2018; Banfi et al., 2025). We propose three complementary ways to quantify the expansion effect that remote work modalities create in the market. First, we pay attention to the *geographic area* covered according to the location of the applicants. Second, we propose a measure of *population covered* by the job ad, by summing up the working age population (18-65) in the municipalities where applicants declare their residence. The two notions may be quite different given the heterogeneous population densities across geographical locations. The population and *trabajando.com*’s applicants are highly concentrated in the central part of the country and even more in the Santiago Metropolitan Region. Therefore, a job ad may expand the population covered much more than the area if its applicants come from densely populated municipalities of Santiago. Finally, an admittedly blunt third measurement for coverage, but it provides a measurement

Table 12: Estimation of equation (5) for log applications received 2010-2024: OLS vs IV.

	(1)	(2)	(3)	(4)	(5)	(6)
WFH prob	0.19*** (0.011)	0.27*** (0.012)	0.83*** (0.030)	1.28*** (0.040)	0.86*** (0.039)	1.98*** (0.075)
HYB prob	0.043*** (0.0037)	0.012*** (0.0039)	0.28*** (0.011)	0.57*** (0.013)	0.16*** (0.014)	0.26*** (0.024)
log wage	0.11*** (0.0015)	0.042*** (0.0016)	0.11*** (0.0015)	0.037*** (0.0016)	-2.33*** (0.022)	-4.14*** (0.045)
Vac Obs	3530465	2652282	3530463	2652280	3530463	2652280
R2	0.22	0.48	-0.00053	-0.011	-0.78	-2.55
Adjusted R2	0.22	0.47	-0.00068	-0.022	-0.78	-2.58
CD test			249396.7	131359.7	9658.8	4065.8
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

for geographic diversity, is the *number of municipalities* declared as residence by applicants of a job ad. Since the three measurements of labor market size yield similar results, we defer the results and discussion of the last two to the appendix A.5.

To quantify the first measurement as “area of coverage” of each job ad, we use a convex hull approach based on the declared municipality, city, or region of residence of each applicant.¹⁰ We search for latitude and longitude of the pragmatic location, typically the capital city or the centroid of the populated portion, for each geographic unit. After filtering outliers with the set of points per each ad, we construct the smallest convex polygon that encompasses the set of geocoded applicant coordinates. In the appendix A.6 we provide a detailed description of the algorithm and a reference illustration of the method is shown in Figure A1.

By calculating the area within this boundary, we derive a measure of the realized catchment area associated with each ad, a geometric proxy of the size of the local labor market.¹¹

¹⁰Most applicants report municipality or county of residence. When this information is missing, we turn to the reported city; if neither is available, we use the declared region. Still, there are a number of missing location of applicants that are excluded from this analysis.

¹¹We exclude job ads with one or two applicants, since the convex hull is ill-defined in these cases. See more details in appendix A.6.

Table 13 shows the results of the estimation of the outcome equation 5 using the logarithm of the coverage area. Both OLS and IV estimates show positive effects, which implies a marginal increase of 0.07-0.21% of the coverage area per each percentage point in which the probability of WFH increases. However, in columns (4) and (6) in which firm fixed effects are included and we instrument for modality, the estimates show a substantial increase of area coverage, increasing by 0.9 log points per percentage point of increase of WFH probability. The difference of IV estimates with and without firm fixed effects is likely driven by excluded firms in columns (4) and (6), single-ad and recruiting firms typically posting less attractive jobs. On the bright side, the IV estimates with fixed effects allow us to isolate the WFH effect from other firm characteristics, potentially confounded with WFH or HYB.

The effect of HYB probability is consistent with a substantial increase in the catchment area when employers advertise in this modality. For IV estimates, the HYB effect is consistently around 0.4 and does not vary much. Due to the nature of WFH jobs, we expected the effect of WFH on the coverage area to be larger. This occurs in all estimates controlling for firm fixed effects (even columns) since the effect of the WFH probability on the catchment area is substantially higher than that of the HYB probability. This suggests that controlling for unobserved employer heterogeneity is relevant to disentangle the effect of remote modalities from other firm features.

The effect of posted log wage in ads mostly affects positively to the coverage area. The estimated impact remains significant and mild when the wage is not instrumented –around 0.02% increase in coverage area after an increase in wage of 1%–, but the effect dramatically changes when wages are instrumented using the D2-average log wage and firm fixed effects are included, i.e. 0.37% job ad coverage increase when posted wage increases 1%.

Applicants’ characteristics: In Table 14 we show the estimates of equation (5) when the dependent variable is the log of the expected wage of applicants (self-declared in their CV). As shown in Banfi et al. (2022), this variable can serve as a proxy of the productivity of the applicant. The WFH and HYB probabilities show a small, sometimes non-significant effect in OLS estimates, showing that this feature is only weakly partially correlated with the wage expectation of applicants. However, the IV estimates of the WFH probability consistently show a negative effect, especially when firm fixed effects are present. For example, according to column (6), an increase of 1% of the probability of WFH of the ad generates a drop of 0.34% of the expected wage of the applicant. In contrast, the IV estimates of the probability of HYB are positive and more moderate in absolute terms than that of WFH, e.g., a 1% increase of the probability of HYB leads to a 0.13% increase of the average expected wage of applicants. These results strongly suggest that the WFH and HYB modalities cause a different sorting effect on applicants when they are moved by exogenous shifts of the

Table 13: Estimation of equation (5) for log area of applicants 2010-2024: OLS vs IV.

	(1)	(2)	(3)	(4)	(5)	(6)
WFH prob	0.15*** 0.01	0.21*** 0.012	0.064** 0.028	0.92*** 0.041	0.067** 0.029	0.86*** 0.041
HYB prob	0.19*** 0.0035	0.090*** 0.0038	0.43*** 0.01	0.36*** 0.013	0.42*** 0.01	0.38*** 0.013
log wage	0.025*** (0.0014)	0.022*** (0.0016)	0.024*** (0.0014)	0.020*** (0.0016)	-0.15*** (0.016)	0.37*** (0.024)
Vac Obs	3821124	2849084	3821116	2849081	3821116	2849081
R2	0.26	0.4	-0.00035	-0.0029	-0.0045	-0.02
Adjusted R2	0.26	0.4	-0.00049	-0.014	-0.0046	-0.031
CD test			255013.1	126247.3	9357.5	4000.5
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

modality probabilities among D2-competitor employers.

The effect of the offered wages is positive, modest and significant when this variable is not instrumented in columns (1) to (4), i.e., around 0.25% increase in the average ' expected wage of the applicants when there is a 1% increase in the offered wage. When the wage is instrumented in columns (5) and (6), the effect becomes substantially larger than in the other columns, with an elasticity of around 1.6 of the average expected wage with respect to the offered wage. This shows that, in addition to the negative response of the applications received estimated in Table 12, and the decrease in the size of the labor market exemplified by the estimates in Tables 13, A11 and A12 when the offered wages increase exogenously, there is also a substantial sorting increase of applications from high-expectation job seekers. These results complete the view that wages act as a way to attract more and more qualified applications in spite of some other application effort being diverted to lower-wage job ads.

In Appendix A.5.2 we also provide Tables A21-A24 in which we estimate model (5) in which the dependent variables are the percentage of applications with college degree for a given ad, and the average experience of applicants. In both cases, the results point in the same direction as those in Table 14. The WFH probability affects negatively in most specifications the percentage of college degree of applicants for the full and post-COVID

Table 14: Estimation of equation (5) for log applicant expected wage 2010-2024: OLS vs IV.

	(1)	(2)	(3)	(4)	(5)	(6)
WFH prob	0.043*** (0.011)	0.019 (0.013)	-0.088*** (0.028)	-0.14*** (0.041)	-0.077** (0.031)	-0.34*** (0.047)
HYB prob	-0.015*** (0.0037)	-0.0039 (0.0043)	0.022** (0.010)	0.042*** (0.013)	0.051*** (0.011)	0.13*** (0.015)
log wage	0.28*** (0.0014)	0.24*** (0.0017)	0.28*** (0.0014)	0.24*** (0.0017)	1.54*** (0.017)	1.66*** (0.028)
Vac Obs	3350632	2523529	3350630	2523527	3350630	2523527
R2	0.23	0.3	0.012	0.0076	-0.23	-0.27
Adjusted R2	0.23	0.29	0.012	-0.0028	-0.23	-0.29
CD test			263219.6	132697.2	9559.1	4008.7
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

sample. Although the sign seems less clear for the case of average experience of applications received for the whole sample, it turns out to be clearly negative after 2020. In contrast, the probability of HYB generates a moderate positive effect on college frequency, as well as on the average experience of applicants. The effect of log offered wage on these quality variables is positive and moderate for specifications in which we do not instrument for wages, but turns to be much higher and positive when we estimate the impact of an exogenous shock through the IV estimation. Therefore, all three variables summarizing the quality of the pool of applicants offer a consistent rationale. The WFH modality induces some negative sorting, while HYB jobs attract slightly better applicants compared to the fully remote modality. Higher wages induce a positive quality sorting according to these three measures of applicant quality.

Robustness: To ensure the stability of our findings, we evaluate whether our results on job competition, labor market size, and applicant quality hold across different time horizons and geographic constraints. We first consider the potential for structural change following the COVID-19 pandemic by re-estimating our models for the 2020–2024 subsample. Despite the large impact of the pandemic on labor market dynamics, the qualitative results for applicant competition and sorting remain remarkably similar to those of the full 2010–2024 period. In

Table A9, we find that the positive impact of WFH and HYB probabilities on the volume of applications persists, with the WFH effect remains substantial: a 1 percentage point increase in WFH probability increases received applications by 1.33% in our benchmark IV specification. Similarly, the sorting effect on applicant quality remains robust in the post-2020 period (Table A19), where WFH probability continues to exert a negative effect on expected wages while HYB probability shows a positive, though more moderate, impact. These findings suggest that the fundamental mechanisms of worker sorting and the “portfolio effect” in application behavior were not fundamentally altered by the pandemic.

Furthermore, we also address potential geographic heterogeneity by restricting our analysis of labor market size to the Metropolitan Region of Santiago (RM). This restriction is important as the RM is the country’s primary labor market and possesses a unique population distribution and density compared to the more sparsely populated regions of Chile. As shown in Tables A14, A16, and A18, the expansionary effect of remote modalities on the labor market remains highly significant within this concentrated geographic area. In the RM-only sample, an increase in the probability of WFH continues to substantially increase the area of the catchment, the population covered, and the number of municipalities served by the reference ad. Interestingly, the magnitude of these effects in the RM is often larger than in the national sample, particularly for population coverage (Table A16), reflecting the high density of the capital and how remote work effectively integrates its various sub-markets. Consistent with our national results, the strategically induced increase in offered wages in the RM leads to a reduction in the number of municipalities covered (Table A18), reinforcing the portfolio logic where high-wage jobs attract localized and intense competition from high-quality applicants at the expense of geographic breadth.

7 Conclusions

The shift toward remote work has fundamentally reshaped the organization of labor, yet the empirical literature has often struggled with the precise measurement of these modalities and their interaction with traditional wage-setting mechanisms. This paper makes a first methodological contribution by developing a multi-stage classification pipeline using a fine-tuned Spanish-language BERT model (BETO) to identify work-from-home (WFH) and hybrid (HYB) job ads from unstructured text. Using the unique features of the *trabajando.com* dataset—specifically the requirement for firms to provide a net monthly wage even when it is hidden from applicants—we circumvent the sample selection biases inherent in most online job board data. This approach allows us to consistently distinguish between fully remote and hybrid arrangements, while ensuring comparability between homogeneous

work-hour arrangements through a simultaneous classification of full-time status.

To address the endogeneity of firm-level decisions, we provide a novel application of network-based social effect models to the labor market. We posit that firms operate in an oligopsonistic environment where wage and modality offerings are strategic reactions to the behavior of competitors. Using the bipartite network structure formed by job applications, we identify local labor markets specific to ads and generate economically meaningful instruments through degree-2 network averages—the characteristics of competitors of a firm’s own competitors. This strategy allows us to isolate exogenously induced shifts in modalities and wages, providing a framework to estimate causal effects under the sensible assumption that the behavior of these distant neighbors affects a focal employer only through its impact on direct competitors.

Our empirical findings reveal that WFH and hybrid work are not just different degrees of the same phenomenon, but distinct strategic choices with distinct impacts on the labor market. Estimation using instruments for modality and strategic interactions shows that fully remote work is associated with a substantial wage premium, suggesting a loss of monopsonic power as firms must compete in a broader, geographically neutral market. In contrast, hybrid arrangements often carry a small wage penalty when instrumented, reflecting their continued geographic constraints and the predominance of amenity value. We also find that remote modalities significantly expand the labor market’s reach, increasing the geographic area, population covered, and number of municipalities from which firms draw talent. However, exogenous wage increases appear to trigger a portfolio effect, where applicants concentrate their efforts on high-return, high-competition positions, thus inducing a sorting effect that increases the average quality and expectations of the applicant pool while potentially narrowing the geographic breadth of applications.

These findings suggest that the integration of Natural Language Processing (NLP) techniques with a structural network approach fundamentally changes our understanding of the impact of remote work on labor market behavior. Moving beyond simple binary classifications and acknowledging the strategic interdependence of firms, we unpack several aspects of wage offering, remote modality choices, and associated competition effects. Although our results are robust to the structural shocks of the COVID-19 pandemic, some caveats remain, such as the potential bias toward larger firms that post frequently due to the use of firm fixed effects. Future research should explore whether these strategic interactions lead to long-term convergence in regional wage disparities or if the persistence of local monopsony in hybrid markets will further widen the gap between high-skilled remote workers and the localized service economy.

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A Appendix

A.1 Dictionary keywords and buckets example

oportunidad de teletrabajo y desde el hogar	modalidad de trabajo a distancia y virtual	Trabajo desde casa con horario flexible	EQUIPO CON OPCION DE TRABAJO EN LINEA Y EN MODALIDAD REMOTA
Colaboración En Línea Y Trabajo A Distancia	Home Based	equipo con opción de trabajo en línea and en modalidad remota	EQUIPO REMOTO Y COLABORACIÓN VIRTUAL
Equipo De Trabajo Con Opción De Ubicación Flexible	Equipo Con Opcion De Horario Remoto And Trabajo En Modalidad Virtual	Trabajo En Modalidad Flexible Y Colaboración Virtual	Posibilidad De Colaboracion En Linea
TRABAJO DESDE CUALQUIER UBICACION, INCLUIDO EL HOGAR	Equipo Con Opcion De Teletrabajo	Equipo con opcion de horario flexible y trabajo en linea	EQUIPO REMOTO CON OPORTUNIDAD DE TELETRABAJO
OPORTUNIDAD DE TRABAJAR EN MODALIDAD REMOTA	modalidad de trabajo desde el hogar en linea	Colaboración en línea y a distancia	equipo con opción de horario remoto and trabajo en modalidad remota

Table A1: Remote work keywords snippet

A.2 Job postings used for test example

Job advertisements used for classifiers test
Asistentes de Venta Planta Lápiz López, importante empresa de Retail, requiere contratar para sus sucursales, a personas que deseen trabajar para el cargo de Asistentes de Venta, donde la función principal es la atención al cliente y la venta de los diversos artículos que se ofrecen en la tienda, además de reposición de productos. La búsqueda está orientada a personas proactivas, motivadas y comprometidas con su trabajo. Tenemos vacantes en las siguientes modalidades: - Mall Costanera Center, Providencia: Part time 20 hrs fines de semana. - Mall Costanera Center, Providencia: Full time 45 hrs semanales de lunes a domingo 5x2. - Mall Apumanque, Las Condes: Full time 45 hrs semanales de lunes a domingo 5x2. - Av El Bosque Norte 119, Providencia: Part time 20 hrs de lunes a jueves de 10:00 a 15:00 hrs. - Outlet La Reina Av. Príncipe de Gales 6853, La Reina: Full time 45 hrs semanales de lunes a sábado 5x2. - Strip Center Av Tobalaba 4507, Full time 45 hrs semanales de lunes a domingo 5x2. - Lider Av Príncipe de Gales 9140, La Reina: Part time 20 hrs viernes de 12:00 a 21:00 hrs, sábado y domingo de 15:00 a 21:00 hrs. - Mall Portal La Reina: Full time 45 hrs semanales de lunes a domingo 5x2. - Av Providencia 2283: Full time 45 hrs semanales de lunes a sábado 5x2. - Espacio Urbano Lo Barnechea: Full time 45 hrs semanales de lunes a domingo 5x2. - Espacio Urbano Lo Barnechea: Part time 20 hrs sábado y domingo. - Espacio Urbano Lo Barnechea: Part time 20 hrs semanales de lunes a viernes de 12:00 a 16:00 hrs. - Mall Plaza Vespucio, La Florida: Full time 45 hrs semanales de lunes a domingo 5x2. - Mall Alto Las Condes: Full time 45 hrs semanales de lunes a domingo 5x2. - Strip Center SubCentro, Las Condes: Full time 45 hrs semanales de lunes a domingo 5x2. - Outlet Vivo La Florida: Part time 20 hrs sábado y domingo. - Outlet Av. Vitacura 6410: Full time 45 hrs semanales de lunes a sábado 5x2. Jornada Completa - Enseñanza media completa - Mayores de 18 años - Disponibilidad para trabajar a largo plazo (EXCLUYENTE) Santiago
Agente de ventas y supervisores Oportunidad de integrarse a prestigiosa empresa multinacional. Horarios flexibles, capacitaciones gratuitas. posibilidad de trabajar desde casa. Sea su propio jefe. Altos ingresos. Part Time personas serias y ambiciosas Stgo.

Table A2: Job postings shown to participants in the WFH choice task snippet

A.3 Description of variable construction

As in [Banfi and Villena-Roldán \(2019\)](#), we construct variables related to whether a job requires some form of additional knowledge. The set of words we use to create these variables are "*exper*", "*conocim*", "*capacit*", which are Spanish roots/stems related to *experience*, *knowledge* and *training* and we use to create a *general knowledge* requirement.

For *specific knowledge* requirements, we search for "*estudio*", "*especiali*", "*dominio*" which relate to *studies*, *specialty* and *mastery/proficiency*.

We use the words "*pc*", "*excel*", "*word*", "*office*", "*power point*", "*ppt*", "*comput*" to categorize job ads in terms of whether they require some basic computer knowledge.

A.4 Additional Figures and Tables

A.4.1 Additional estimates of the log wage equation (2) estimates

Table A3: Estimation of log wage equation (2) 2020-2024: OLS vs IV estimates

	(1)	(2)	(3)	(4)	(5)	(6)
WFH prob	0.048*** (0.0042)	0.027*** (0.0057)	0.14*** (0.012)	0.27*** (0.022)	0.11*** (0.013)	0.23*** (0.022)
HYB prob	0.053*** (0.0013)	0.040*** (0.0015)	-0.053*** (0.0047)	-0.096*** (0.0064)	-0.064*** (0.0048)	-0.11*** (0.0064)
D1-avg log wage	0.62*** (0.0014)	0.54*** (0.0016)	0.62*** (0.0014)	0.55*** (0.0016)	0.48*** (0.019)	0.41*** (0.016)
Vac Obs	1441033	1025351	1438040	1023345	1438040	1023345
R2	0.59	0.69	0.12	0.093	0.11	0.085
Adjusted R2	0.59	0.69	0.12	0.08	0.11	0.073
CD test			57783.1	30450.7	2429.6	3689.4
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: monthly seasonal dummies and linear trend.

Table A4: Estimation of log wage equation (2) 2010-2024: OLS vs IV, including market size

	(1)	(2)	(3)	(4)	(5)	(6)
WFH prob	0.046*** (0.0032)	0.032*** (0.0039)	0.17*** (0.0095)	0.27*** (0.015)	0.15*** (0.0096)	0.24*** (0.016)
HYB prob	0.031*** (0.0011)	0.032*** (0.0013)	-0.026*** (0.0035)	-0.022*** (0.0048)	-0.042*** (0.0036)	-0.038*** (0.0048)
D1-avg log wage	0.76*** (0.00075)	0.64*** (0.00091)	0.76*** (0.00075)	0.64*** (0.00091)	0.57*** (0.0086)	0.45*** (0.011)
log sum D1 vacs	0.0092*** (0.00010)	0.0057*** (0.00013)	0.0070*** (0.00011)	0.0023*** (0.00014)	0.0064*** (0.00011)	0.0017*** (0.00015)
Vac Obs	4564115	3344093	4558817	3340526	4558817	3340526
R2	0.67	0.75	0.18	0.13	0.17	0.12
Adjusted R2	0.67	0.74	0.18	0.12	0.17	0.11
CD test			158479.2	73684	8759.7	5697.6
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: monthly seasonal dummies and linear trend.

A.4.2 Additional estimates of the WFH prob. equation (3) estimates

Table A5: Estimation of WFH prob equation (3) 2020-2024: OLS vs IV estimates

	(1)	(2)	(3)	(4)	(5)	(6)
log wage	0.00084*** (0.00015)	0.00037** (0.00016)	-0.00018 (0.00016)	-0.0001 (0.00017)	-0.026*** (0.0080)	0.027*** (0.0066)
D1-avg WFH prob	0.65*** (0.0019)	0.33*** (0.0022)	1.60*** (0.0039)	0.87*** (0.0034)	1.60*** (0.0040)	0.87*** (0.0036)
D1-avg HYB prob	0.0043*** (0.00056)	0.017*** (0.00060)	-0.056*** (0.0012)	-0.022*** (0.0012)	-0.060*** (0.0015)	-0.018*** (0.0015)
Vac Obs	1441033	1025351	1438040	1023345	1438040	1023345
R2	0.16	0.31	-0.081	-0.034	-0.1	-0.06
Adjusted R2	0.16	0.3	-0.081	-0.047	-0.1	-0.074
CD test			238905.6	161927.4	195	221.4
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

Table A6: Estimation of WFH equation (3) 2010-2024: OLS vs IV, including market size

	(1)	(2)	(3)	(4)	(5)	(6)
log wage	0.00089*** (0.000061)	0.00052*** (0.000072)	0.00026*** (0.000065)	0.00018** (0.000075)	-0.012*** (0.0023)	0.0017 (0.0035)
D1-avg WFH prob	0.71*** (0.0013)	0.42*** (0.0015)	1.75*** (0.0026)	1.16*** (0.0026)	1.76*** (0.0026)	1.16*** (0.0027)
D1-avg HYB prob	0.022*** (0.00034)	0.029*** (0.00040)	-0.053*** (0.00073)	-0.030*** (0.00086)	-0.054*** (0.00077)	-0.030*** (0.00091)
log sum D1 vacs	-0.00010*** (0.000015)	0.00019*** (0.000019)	-0.00010*** (0.000017)	0.00017*** (0.000021)	-0.000047** (0.000019)	0.00017*** (0.000021)
Vac Obs	4564115	3344093	4558817	3340526	4558817	3340526
R2	0.11	0.22	-0.066	-0.042	-0.074	-0.042
Adjusted R2	0.11	0.21	-0.066	-0.053	-0.075	-0.053
CD test			485311.7	322122.5	930.5	376
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

A.4.3 Additional estimates of the HYB prob. equation (4) estimates

Table A7: Estimation of HYB prob equation (4) 2020-2024: OLS vs IV.

	(1)	(2)	(3)	(4)	(5)	(6)
log wage	0.014*** (0.00048)	0.011*** (0.00059)	0.0082*** (0.00049)	0.0068*** (0.00060)	-0.33*** (0.028)	-0.26*** (0.025)
D1-avg WFH prob	0.018*** (0.0062)	0.031*** (0.0080)	-0.32*** (0.012)	-0.25*** (0.012)	-0.31*** (0.014)	-0.22*** (0.014)
D1-avg HYB prob	0.79*** (0.0018)	0.71*** (0.0021)	1.25*** (0.0037)	1.10*** (0.0044)	1.21*** (0.0052)	1.05*** (0.0058)
Vac Obs	1441033	1025351	1438040	1023345	1438040	1023345
R2	0.34	0.52	0.082	0.074	-0.22	-0.11
Adjusted R2	0.34	0.51	0.082	0.061	-0.22	-0.12
CD test			238905.6	161927.4	195	221.4
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

Table A8: Estimation of HYB equation (4) 2010-2024: OLS vs IV, including market size

	(1)	(2)	(3)	(4)	(5)	(6)
log wage	0.0033*** (0.00016)	0.0037*** (0.00020)	0.0038*** (0.00017)	0.0030*** (0.00021)	-0.11*** (0.0061)	-0.074*** (0.0098)
D1-avg WFH prob	0.10*** (0.0034)	0.053*** (0.0043)	-0.20*** (0.0066)	-0.22*** (0.0072)	-0.19*** (0.0070)	-0.21*** (0.0076)
D1-avg HYB prob	0.95*** (0.00090)	0.89*** (0.0011)	1.33*** (0.0019)	1.23*** (0.0024)	1.32*** (0.0020)	1.22*** (0.0025)
log sum D1 vacs	0.00050*** (0.000039)	-5.8e-05 (0.000052)	-0.00021*** (0.000042)	-0.00036*** (0.000057)	0.00034*** (0.000052)	-0.00033*** (0.000058)
Vac Obs	4564115	3344093	4558817	3340526	4558817	3340526
R2	0.33	0.49	0.18	0.15	0.09	0.11
Adjusted R2	0.33	0.48	0.18	0.14	0.089	0.1
CD test			485311.7	322122.5	930.5	376
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

A.4.4 Additional estimates of log applications

Table A9: Estimation of equation (5) for log applications 2020-2024: OLS vs IV.

	(1)	(2)	(3)	(4)	(5)	(6)
WFH prob	0.15*** (0.012)	0.27*** (0.014)	0.41*** (0.030)	0.69*** (0.045)	0.21*** (0.040)	1.33*** (0.094)
HYB prob	-0.0068** (0.0034)	-0.00041 (0.0038)	0.14*** (0.011)	0.34*** (0.014)	0.0037 (0.014)	-0.20*** (0.029)
log wage	0.052*** (0.0022)	-0.022*** (0.0025)	0.048*** (0.0022)	-0.030*** (0.0025)	-1.96*** (0.039)	-3.95*** (0.098)
Vac Obs	1061192	804111	1061192	804111	1061192	804111
R2	0.28	0.47	-0.0016	-0.011	-0.78	-3.23
Adjusted R2	0.28	0.47	-0.0021	-0.023	-0.78	-3.28
CD test			58795.1	32649.1	1985.5	704.5
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

Table A10: Estimation of equation (5) for log applications 2010-2024: OLS vs IV, including market size

	(1)	(2)	(3)	(4)	(5)	(6)
WFH prob	0.18*** (0.0067)	0.17*** (0.0074)	0.64*** (0.018)	0.71*** (0.025)	0.61*** (0.035)	0.16*** (0.055)
HYB prob	-0.064*** (0.0022)	-0.053*** (0.0024)	-0.27*** (0.0064)	-0.16*** (0.0081)	-0.12*** (0.012)	0.070*** (0.018)
log wage	0.079*** (0.00088)	0.045*** (0.0010)	0.076*** (0.00088)	0.045*** (0.0010)	2.88*** (0.018)	3.28*** (0.033)
log sum D1 vacs	0.69*** (0.00027)	0.65*** (0.00032)	0.75*** (0.00029)	0.71*** (0.00034)	0.73*** (0.00058)	0.72*** (0.00077)
Vac Obs	3530465	2652282	3530463	2652280	3530463	2652280
R2	0.72	0.8	0.64	0.61	-0.38	-0.91
Adjusted R2	0.72	0.8	0.64	0.61	-0.38	-0.93
CD test			165921.7	87581.9	8160.2	3074
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

A.5 Additional measurements of labor market size

Turning to the population covered by the job ad, we show the estimates in Table A11. The overall results seem to be quite aligned with the measurement area covered. Both OLS estimates increase in a sizable way the population covered when the WFH or HYB probability increases. However, the effect magnifies substantially when we instrument for modalities in columns (3) to (6). The largest estimates show that an increase in the probability of WFH of 1% expands the population covered by 2.4%, provided the modality shift had been triggered by the WFH/HYB modality of D2-competitors. The gap between WFH and HYB modality is especially large when we control for firm fixed effects, again suggesting that employer specific amenities may be confounded with remote modalities presence.

The effect of wages on the covered population turns out to be moderate and positive when we do not instrument this variable. When we do, in columns (5) and (6), the marginal effects become large and negative. We interpret that OLS estimates show the partial correlation between more attractive, high-wage jobs and the population covered. The IV estimates instead reveal the effect triggered by an increase in the wage of the D2 employer competitors. When the latter increase their wages, they induce a negative reaction of the reference employer’s wage (see first-stages in Appendix ABC), which suggests strategic substitutability in pricing strategies, an effect consistent with results for application in equation Table 12. Although these results seem at odds with predictions in standard directed search models, it naturally arises when we consider multiple simultaneous applications that occur typically in online job boards and risk-averse workers, as in Galenianos and Kircher (2009). In this setting a “portfolio effect” arises: when a competitor exogenously increases its wage, they attract more applicants, but the increased competition makes their job a riskier bet. The optimal response for the reference employer is to lower their wage to serve as the “insurance” application in the worker’s portfolio.

The third measurement is the simple count of the municipalities in which applicants to a given job ad reside. The estimates of modalities and wages are shown in Table A12. The results align well with those shown in Tables 13 and A11. Once again, the OLS results show positive but low responses to the modalities and wages, as for the other measurements. IV estimates, especially when firm fixed effects are present in columns (4) and (6), are substantially larger. An increase of 1% of the probability of WFH increases the number of municipalities covered by the ad to 0.17 or 0.27 in columns (4) and (6). The effect is positive and significant for the probability of HYB, but it is substantially lower than that of WFH. These estimates not only quantify a larger coverage but also a geographic diversity increase due to the exogenously induced modality change of job ads. As in the case of population coverage, the effect of an exogenous change in the reference ad wage

Table A11: Estimation of equation (5) for log population covered 2010-2024: OLS vs IV.

	(1)	(2)	(3)	(4)	(5)	(6)
WFH prob	0.21*** (0.01)	0.30*** (0.011)	0.87*** (0.03)	1.58*** (0.045)	1.08*** (0.041)	2.38*** (0.11)
HYB prob	0.27*** (0.0035)	0.19*** (0.0037)	0.66*** (0.011)	0.93*** (0.014)	0.44*** (0.016)	0.49*** (0.033)
log wage	0.12*** (0.0013)	0.059*** (0.0015)	0.12*** (0.0013)	0.053*** (0.0015)	-2.44*** (0.066)	-5.51*** (0.16)
Vac Obs	4600617	3369563	4563697	3344417	4563697	3344417
R2	0.2	0.46	-0.00088	-0.016	-0.8	-4.35
Adjusted R2	0.2	0.45	-0.001	-0.027	-0.8	-4.4
CD test			236511.1	110766.1	1116.8	489.7
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

due to a change in the D2 wage of competitors defies the basic intuition built in models with complementary strategic competition. Again, the portfolio logic in the [Galenianos and Kircher \(2009\)](#) model helps rationalize the striking result: a 1% exogenously triggered increase in wages reduces the expected number of municipalities covered by the ad by 0.34-0.56 depending on the introduction of firm fixed effects. Essentially, firms that increase wages reduce the number of municipalities due to the increased competition they create for the position and the competitive reaction of wage reduction they induce in other employers.

A.5.1 Additional estimates of labor market size measurements

Table A12: Estimation of equation (5) for number municipalities covered 2010-2024: OLS vs IV.

	(1)	(2)	(3)	(4)	(5)	(6)
WFH prob	3.01*** (0.15)	3.41*** (0.17)	12.0*** (0.39)	17.2*** (0.56)	12.5*** (0.55)	26.8*** (1.03)
HYB prob	1.44*** (0.049)	1.21*** (0.055)	4.81*** (0.14)	8.46*** (0.18)	2.98*** (0.20)	4.20*** (0.33)
log wage	1.80*** (0.019)	0.42*** (0.023)	1.77*** (0.019)	0.36*** (0.023)	-34.7*** (0.30)	-56.5*** (0.61)
Vac Obs	3530465	2652282	3530463	2652280	3530463	2652280
R2	0.19	0.43	0.00024	-0.0095	-0.99	-2.42
Adjusted R2	0.19	0.42	9.3e-05	-0.02	-0.99	-2.45
CD test			249396.7	131359.7	9658.8	4065.8
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

Table A13: Estimation of equation (5) for log area of applicants 2020-2024: OLS vs IV.

	(1)	(2)	(3)	(4)	(5)	(6)
WFH prob	-0.086*** 0.011	0.037*** 0.011	-0.60*** 0.028	0.19*** 0.037	-0.63*** 0.028	0.32*** 0.042
HYB prob	0.17*** 0.0031	0.085*** 0.003	0.24*** 0.01	0.17*** 0.011	0.22*** 0.01	0.068*** 0.013
log wage	-0.019*** 0.002	-0.023*** 0.0019	-0.020*** 0.002	-0.024*** 0.0019	-0.33*** 0.029	-0.91*** 0.045
Vac Obs	1167395	869408	1167395	869408	1167395	869408
R2	0.16	0.34	6.6e-05	-0.00019	-0.021	-0.25
Adjusted R2	0.16	0.33	-0.00038	-0.012	-0.021	-0.27
CD test			61491.3	33712.1	1810.8	641.8
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

Table A14: Estimation of equation (5) for log area of applicants 2010-2024: OLS vs IV, only RM

	(1)	(2)	(3)	(4)	(5)	(6)
WFH prob	0.14*** 0.018	0.13*** 0.019	0.67*** 0.066	0.98*** 0.071	0.67*** 0.079	1.39*** 0.16
HYB prob	0.18*** 0.0053	0.15*** 0.0056	0.36*** 0.015	0.36*** 0.016	0.48*** 0.018	0.61*** 0.038
log wage	0.012*** 0.0018	-0.0079*** 0.0021	0.011*** 0.0018	-0.0094*** 0.0021	1.76*** 0.092	5.23*** 0.38
Vac Obs	2165963	1721968	2150627	1710672	2150627	1710672
R2	0.29	0.42	-0.00038	-0.0018	-0.42	-3.65
Adjusted R2	0.29	0.41	-0.00062	-0.012	-0.42	-3.7
CD test			88454.4	67728.7	410.9	81.2
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

Table A15: Estimation of equation (5) for log population covered 2020-2024: OLS vs IV.

	(1)	(2)	(3)	(4)	(5)	(6)
WFH prob	0.11*** (0.012)	0.30*** (0.015)	0.47*** (0.034)	1.23*** (0.06)	0.47*** (0.04)	1.38*** (0.069)
HYB prob	0.12*** (0.0037)	0.13*** (0.0041)	0.50*** (0.013)	0.74*** (0.017)	0.65*** (0.02)	0.54*** (0.025)
log wage	0.078*** (0.0022)	0.0031 (0.0026)	0.069*** (0.0023)	-0.011*** (0.0026)	1.68*** (0.14)	-1.37*** (0.12)
Vac Obs	1460299	1038251	1441049	1025732	1441049	1025732
R2	0.26	0.46	-0.0066	-0.025	-0.38	-0.3
Adjusted R2	0.26	0.45	-0.007	-0.039	-0.38	-0.31
CD test			57534	30763.9	178.8	202.9
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

Table A16: Estimation of equation (5) for log population covered 2010-2024: OLS vs IV, only RM

	(1)	(2)	(3)	(4)	(5)	(6)
WFH prob	0.17*** (0.02)	0.27*** (0.02)	0.81*** (0.072)	1.85*** (0.073)	0.81*** (0.1)	1.18*** (0.24)
HYB prob	0.28*** (0.0058)	0.22*** (0.0057)	0.83*** (0.016)	1.03*** (0.016)	0.63*** (0.024)	0.64*** (0.058)
log wage	0.17*** (0.002)	0.067*** (0.0022)	0.16*** (0.002)	0.065*** (0.0022)	-2.89*** (0.12)	-8.58*** (0.58)
Vac Obs	2165246	1721335	2150023	1710137	2150023	1710137
R2	0.21	0.45	-0.00085	-0.016	-1.09	-9.49
Adjusted R2	0.21	0.44	-0.0011	-0.026	-1.09	-9.6
CD test			88474.7	67715.5	412	81.8
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

Table A17: Estimation of equation (5) for number of municipalities covered 2020-2024: OLS vs IV.

	(1)	(2)	(3)	(4)	(5)	(6)
WFH prob	3.01*** (0.13)	3.79*** (0.16)	7.17*** (0.32)	9.97*** (0.52)	5.06*** (0.42)	16.5*** (0.99)
HYB prob	0.53*** (0.037)	0.73*** (0.043)	2.85*** (0.12)	5.20*** (0.16)	1.41*** (0.15)	-0.22 (0.31)
log wage	1.09*** (0.024)	0.076*** (0.028)	1.02*** (0.024)	-0.02 (0.028)	-19.5*** (0.42)	-39.8*** (1.02)
Vac Obs	1061192	804111	1061192	804111	1061192	804111
R2	0.24	0.43	-0.002	-0.015	-0.69	-2.57
Adjusted R2	0.24	0.42	-0.0025	-0.027	-0.69	-2.61
CD test			58795.1	32649.1	1985.5	704.5
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

Table A18: Estimation of equation (5) for number of municipalities covered 2010-2024: OLS vs IV, only RM

	(1)	(2)	(3)	(4)	(5)	(6)
WFH prob	1.91*** (0.24)	3.31*** (0.24)	11.9*** (0.89)	25.0*** (0.90)	12.0*** (1.38)	15.9*** (3.16)
HYB prob	1.88*** (0.071)	0.99*** (0.070)	7.70*** (0.20)	10.6*** (0.20)	4.89*** (0.32)	5.29*** (0.77)
log wage	2.23*** (0.024)	0.62*** (0.026)	2.20*** (0.025)	0.59*** (0.027)	-40.5*** (1.59)	-115.1*** (7.74)
Vac Obs	2165960	1721965	2150626	1710671	2150626	1710671
R2	0.21	0.48	-8.4e-05	-0.017	-1.4	-11.2
Adjusted R2	0.21	0.47	-0.00033	-0.027	-1.4	-11.3
CD test			88454.3	67728.7	410.9	81.2
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

A.5.2 Additional estimates for applicant' quality

Table A19: Estimation of equation (5) for applicants' log expected wage 2020-2024: OLS vs IV.

	(1)	(2)	(3)	(4)	(5)	(6)
WFH prob	-0.00057 (0.018)	-0.0075 (0.023)	-0.40*** (0.044)	-0.38*** (0.068)	-0.21*** (0.050)	-0.69*** (0.083)
HYB prob	-0.0093* (0.0052)	0.0018 (0.0062)	-0.075*** (0.015)	-0.019 (0.021)	0.014 (0.018)	0.25*** (0.026)
log wage	0.23*** (0.0033)	0.19*** (0.0039)	0.24*** (0.0033)	0.19*** (0.0040)	2.00*** (0.049)	2.40*** (0.089)
Vac Obs	903592	691435	903592	691435	903592	691435
R2	0.16	0.22	0.005	0.0031	-0.32	-0.46
Adjusted R2	0.16	0.21	0.0044	-0.0079	-0.32	-0.48
CD test			56020.4	32299.2	1744.6	645.4
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

Table A20: Estimation of equation (5) for applicants' log expected wage 2010-2024: OLS vs IV, including market size

	(1)	(2)	(3)	(4)	(5)	(6)
WFH prob	0.042*** (0.011)	0.017 (0.013)	-0.090*** (0.028)	-0.15*** (0.041)	-0.076** (0.031)	-0.36*** (0.047)
HYB prob	-0.016*** (0.0037)	-0.0049 (0.0043)	0.020* (0.010)	0.036*** (0.013)	0.052*** (0.011)	0.12*** (0.015)
log wage	0.28*** (0.0014)	0.24*** (0.0017)	0.28*** (0.0014)	0.24*** (0.0017)	1.53*** (0.016)	1.77*** (0.028)
log sum D1 vacs	0.0045*** (0.00043)	0.0087*** (0.00054)	0.0036*** (0.00046)	0.0062*** (0.00058)	-0.0027*** (0.00052)	0.010*** (0.00067)
Vac Obs	3350632	2523529	3350630	2523527	3350630	2523527
R2	0.23	0.3	0.012	0.0077	-0.23	-0.32
Adjusted R2	0.23	0.29	0.012	-0.0027	-0.23	-0.33
CD test			175206.1	88373.9	8081.1	3045.9
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

Table A21: Estimation of equation (5) for average college of applicants 2010-2024: OLS vs IV.

	(1)	(2)	(3)	(4)	(5)	(6)
WFH prob	0.0034** (0.0017)	-0.0070*** (0.0020)	-0.042*** (0.0045)	-0.11*** (0.0066)	-0.047*** (0.0062)	-0.19*** (0.0097)
HYB prob	0.010*** (0.00058)	0.017*** (0.00065)	0.017*** (0.0016)	0.0078*** (0.0022)	0.037*** (0.0022)	0.042*** (0.0031)
log wage	0.079*** (0.00023)	0.068*** (0.00027)	0.079*** (0.00023)	0.069*** (0.00027)	0.48*** (0.0035)	0.53*** (0.0058)
Vac Obs	3530465	2652282	3530463	2652280	3530463	2652280
R2	0.68	0.73	0.033	0.023	-0.84	-1.07
Adjusted R2	0.68	0.73	0.033	0.013	-0.84	-1.09
CD test			249396.7	131359.7	9658.8	4065.8
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

Table A22: Estimation of equation (5) for average college of applicants 2020-2024: OLS vs IV.

	(1)	(2)	(3)	(4)	(5)	(6)
WFH prob	0.017*** (0.0022)	0.0037 (0.0028)	-0.015*** (0.0056)	-0.055*** (0.0089)	0.058*** (0.011)	-0.21*** (0.021)
HYB prob	0.022*** (0.00064)	0.023*** (0.00075)	0.030*** (0.0020)	0.013*** (0.0027)	0.080*** (0.0039)	0.14*** (0.0066)
log wage	0.059*** (0.00041)	0.050*** (0.00049)	0.059*** (0.00042)	0.050*** (0.00049)	0.77*** (0.011)	0.97*** (0.022)
Vac Obs	1061192	804111	1061192	804111	1061192	804111
R2	0.69	0.73	0.02	0.013	-2.75	-4.43
Adjusted R2	0.69	0.73	0.02	0.0019	-2.75	-4.5
CD test			58795.1	32649.1	1985.5	704.5
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

Table A23: Estimation of equation (5) for average experience of applicants 2010-2024: OLS vs IV.

	(1)	(2)	(3)	(4)	(5)	(6)
WFH prob	0.0074 (0.028)	-0.15*** (0.032)	0.66*** (0.075)	0.12 (0.10)	0.56*** (0.12)	-1.43*** (0.18)
HYB prob	0.13*** (0.0095)	0.018* (0.010)	0.19*** (0.027)	0.32*** (0.034)	0.62*** (0.042)	1.01*** (0.057)
log wage	1.25*** (0.0037)	0.96*** (0.0043)	1.25*** (0.0037)	0.95*** (0.0043)	9.76*** (0.065)	10.2*** (0.10)
Vac Obs	3530144	2652067	3530142	2652065	3530142	2652065
R2	0.45	0.58	0.031	0.018	-1.41	-1.74
Adjusted R2	0.45	0.58	0.031	0.008	-1.41	-1.77
CD test			249377.4	131346.8	9675.5	4068.4
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

Table A24: Estimation of equation (5) for average experience of applicants 2020-2024: OLS vs IV.

	(1)	(2)	(3)	(4)	(5)	(6)
WFH prob	-0.79*** (0.040)	-0.50*** (0.049)	-1.31*** (0.099)	-1.40*** (0.15)	-0.13 (0.18)	-3.45*** (0.31)
HYB prob	-0.21*** (0.011)	-0.11*** (0.013)	-0.61*** (0.036)	-0.24*** (0.047)	0.19*** (0.065)	1.46*** (0.095)
log wage	1.09*** (0.0073)	0.75*** (0.0083)	1.10*** (0.0074)	0.76*** (0.0084)	12.5*** (0.18)	13.3*** (0.32)
Vac Obs	1060871	803896	1060871	803896	1060871	803896
R2	0.44	0.57	0.019	0.0098	-2.25	-2.83
Adjusted R2	0.44	0.57	0.019	-0.0018	-2.25	-2.88
CD test			58780	32644.6	1995	706.9
Ind FE	Y	Y	Y	Y	Y	Y
Occ FE	Y	Y	Y	Y	Y	Y
Job Title FE	Y	Y	Y	Y	Y	Y
Firm FE	N	Y	N	Y	N	Y
Method	OLS	OLS	IV mod	IV mod	IV all	IV all

Note: Standard errors in parentheses. * $p < .1$ ** $p < .05$ *** $p < .01$. Additional controls: year and monthly seasonal effects.

A.6 Geographic Catchment Area Estimation

To quantify the spatial reach of the job ads, we implement a robust geometric algorithm that estimates the realized catchment area (in km^2) based on the geographic distribution of the applicants. The methodology employs a hybrid approach that adaptively scales based on the number of applicants per ad (N) to account for dimensionality constraints in small-sample ads

A.6.1 Computational Environment

The algorithm is executed within a **Stata 16+** environment utilizing the `python:` integration block. Spatial computations are performed using the following Python libraries:

- `scipy.spatial`: Implementation of the Quickhull algorithm for convex hull construction.
- `pyproj`: Management of dynamic azimuthal equidistant projections to ensure area-preserving transformations.
- `numpy` and `pandas`: High-performance vectorization and data manipulation for large-scale datasets.

For ads with three or more applicants, we construct the smallest convex polygon that encloses the applicant set. The process follows these steps:

1. **Outlier Mitigation:** To prevent idiosyncratic search behavior from overstating the market size, we filter the furthest 5% of applicants from the centroid.
2. **Dynamic Projection:** Coordinates are projected using an *Azimuthal Equidistant Projection* centered on the median coordinates of the applicant set to minimize distortion.
3. **Calculation:** The area is computed using the Quickhull algorithm [Barber et al. \(1996\)](#). If the resulting point set is collinear, the algorithm defaults to the $N = 1$ circular buffer to ensure continuity.

Figure A1: Illustration of convex hull method for a job ad area coverage

